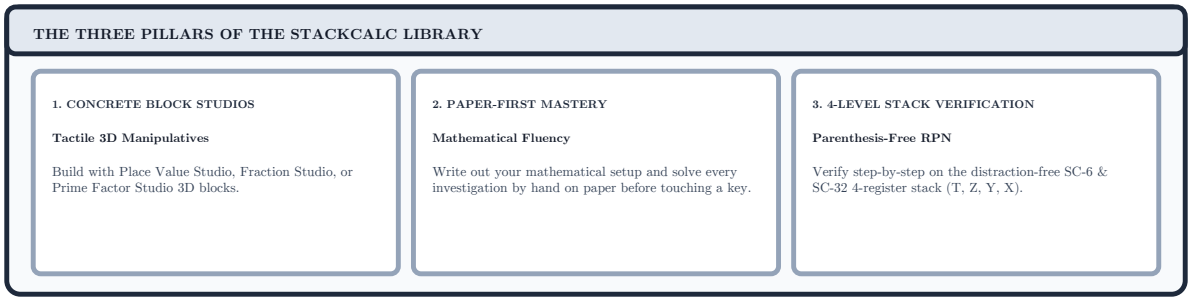
THE STACKCALC LIBRARY • CURRICULUM SERIES — BOOK 1

TEACHING WITH THE STACK

A Practical Concept-First Teacher & Homeschool Guide to Hands-On Mathematics (Grades K–12)

A self-contained, open-and-go K–12 teacher and homeschool parent guide connecting foundational mathematical concepts with our 3 Published Physical Block Studios (Place Value, Fraction & Prime Factor Studios), paper-and-pencil mastery, and distraction-free verification on the four-line StackCalc SC-6 and SC-32 calculators. Works seamlessly alongside any homeschool curriculum (Singapore Math, Saxon, RightStart, Math-U-See, Beast Academy/AoPS) or state-mandated school textbook (Eureka, Illustrative, enVision, Big Ideas, Bridges) with zero prep.



StackCalc

stackcalc.io

First Edition

BOOKS, BLOCKS & INSTRUMENTS IN THE STACKCALC LIBRARY

Illustrated Companion Trade Books (History of Mathematics & Stack Thinking)

- Volume I — From Blocks to the Stack: How Hands-On Blocks Make Numbers Click (Grades K–6 / StackCalc SC-6)
- Volume II — Thinking in the Stack: The Art of Solving Equations from the Inside Out (Grades 7–12, AP & Engineers / StackCalc SC-32)

K–12 Concept-First Curriculum & Grade-Band Student Workbooks (US Letter 8.5" × 11" Editions)

- Book 1 — Teaching with the Stack: Unified K–12 Master Teacher's Guide, Concept Facilitation & Coaching Notes
- Book 2A — Early Elementary Math Lab: Grades K–2 SC-6 Student Workbook (10 Eleven-Page Units + 4 Exams • 840 Problems • 128 Pages)
- Book 2B — Upper Elementary Math Lab: Grades 3–5 SC-6 Student Workbook (10 Eleven-Page Units + 4 Exams • 840 Problems • 128 Pages)
- Book 3 — Middle School Math Lab: Grades 6–8 Two-Method SC-6 Student Workbook (10 Units • 1,000 Problems • 128 Pages)
- Book 4 — High School & STEM Lab: Grades 9–12 & AP SC-32 Student Workbook (10 Eleven-Page Units + 4 Exams • 840 Problems • 128 Pages)
- Book 5 — Stack Speed & Fluency: Full-Year Speed Lessons, Playbooks & 180 Progressive Practice Tests (3,690 Problems • 128 Pages)
- Complete Online Answer Keys & Keystroke Verification Portal: https://stackcalc.io/curriculum/answers

Three Published Physical Manipulative Block Studios & Four-Line Stack Calculators (stackcalc.io)

- Place Value Studio: 6×2 Whole & Decimal Trays, Digit Blocks (0–9) & Stackable Gold [+1] Carry Tokens
- Fraction Studio: 84.0 mm Proportional Fraction Towers (1 Whole to 1/12) & 12-Socket Comparator Lid
- Prime Factor Studio: Logarithmic Prime Blocks (2, 3, 5, 7 where equal products stand flush) & GCF/LCM Comparator
- StackCalc Calculators: SC-6 Elementary/Middle School Calculator & SC-32 Scientific Calculator (Hardware, iOS, iPadOS, watchOS & Web)

Automated Unit-Test Verification & 36-Week Open-Education Pacing Architecture

- 100% Unit-Tested Problems: Every mathematical solution and RPN keystroke sequence across the curriculum is verified by automated Python and Swift RPNCore unit tests.
- Flexible Pacing: Compatible with Singapore Math, Saxon, RightStart, Beast Academy/AoPS, Eureka, Illustrative Math, Big Ideas, and Bridges.

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TEACHING WITH THE STACK: A Practical Concept-First Teacher & Homeschool Guide to Hands-On Mathematics (Grades K–12)

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State & District Curriculum Alignment:

Designed to supplement state-mandated mathematics standards—including Common Core State Standards (CCSS-M), Texas Essential Knowledge and Skills (TEKS), Florida B.E.S.T., Virginia SOL, and AP Mathematics / Computer Science—alongside district basal textbooks (Eureka Math / EngageNY, Illustrative Mathematics / Open Up, enVision Mathematics, Big Ideas Math, HMH Into Math, McGraw-Hill Reveal, Bridges in Mathematics, and Singapore Dimensions Math).

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Physical calculator instruments that utilize coin or button cell batteries must maintain a captive, tool-secured battery compartment compliant with UL 4200A. 3D-printed manipulative blocks constitute small parts under 16 CFR Part 1501; adult supervision is required for children under 6 years of age.

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Preface: Why Teach Mathematics with Blocks & a Stack?

How tactile blocks, paper-first problem solving, and a distraction-free four-line stack build deep mathematical thinkers.

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1. WHY PARENTS & TEACHERS RIGHTLY WORRY ABOUT 'TECHNOLOGY' IN MATH

If you are a homeschooling parent or classroom teacher who is protective of mental math, paper-and-pencil grit, and screen-free childhoods, your skepticism toward 'math technology' is 100% justified:

• Glowing Screens & Gamified Apps Fragment Attention: Tablets, web apps, and cartoon math games train children to guess at multiple-choice buttons rather than patiently working through problems on paper.

• Standard [=] Calculators Become a Mental Crutch: When handed a conventional 1-line calculator with an [=] key, students stop thinking about structure. They type long strings of buttons from left to right—e.g., (8 + 4) ÷ (9 − 3) =—and copy whatever decimal appears, eroding their number sense and order of operations.

2. WHY THE STACK IS NEVER A CRUTCH — IT IS AN ABACUS FOR ORDER OF OPERATIONS

Our program was created as the antidote to both screen addiction and calculator crutches. First, the StackCalc SC-6 and SC-32 are quiet, single-purpose physical instruments—like an abacus, a slide rule, or a compass—with zero internet, zero apps, zero games, and zero notifications. Second, students NEVER use the calculator to skip paper-and-pencil math: in every workbook, the rule is Solve by Hand First, Then Verify on the Stack.

Why verify on a 4-line Stack (T, Z, Y, X) with an [ENTER] key and no [=] or parentheses keys? Because on paper, you write the first number on a top Waiting Line (Line Y), write the second number below it on your Workbench (Line X), and then apply the operation (+, −, ×, ÷). Because there are no parentheses keys, a child cannot use the stack unless they ALREADY understand order of operations: to check $(8 + 4) \div (9 - 3)$, they must evaluate $8 + 4 = 12$ first, let 12 wait visibly on Line Y while computing $9 - 3 = 6$ on Line X, and divide 12 by 6. It reinforces paper math instead of replacing it!

3. HOW EVERY LESSON HELPS STUDENTS FLOURISH: 1. CONCEPT → 2. BLOCKS → 3. PAPER → 4. CHECK

Step 1 — Understand the Mathematical Concept (100% Self-Contained, Zero Prep Required)

Every unit begins with a clear, self-contained explanation of why the math works—how a quantity is composed, regrouped, or factored—along with a word-for-word 'What to Say & Ask' script so any parent or teacher can guide the lesson with confidence.

Step 2 — Touch & Build with Our 3 Physical Block Studios (Concrete Before Abstract)

Before memorizing any rule, students build Place Value, Fraction, and Prime Factor concepts with our 3 Published Physical Block Studios—Place Value Studio (0–9 digit blocks & [+1] carry tokens on a 2-row tray mirroring Lines Y and X), Fraction Studio (84 mm proportional fraction towers where equivalent fractions stand dead-flush), and Prime Factor Studio (logarithmic prime blocks 2, 3, 5, 7 where $h(p) = 7.0 \cdot \log_2(p)$ mm)—before solving by hand in their workbook.

Step 3 — Solve by Hand on Paper First, Then Self-Check on the Stack (Independent Learning)

Students write out their mathematical setup and solve by hand in their workbook first. Then they verify their answer on the SC-6 (Grades K–8) or SC-32 (Grades 8–12)—getting immediate, honest feedback at the kitchen table or desk without peeking at a back-of-book answer key!

Keep the Curriculum You Love — How to Plug This In

How homeschooling families and classroom teachers use these self-contained lessons alongside any math program.

1. NO CURRICULUM-HOPPING REQUIRED: WORKS WITH SINGAPORE, SAXON, AOPS OR DISTRICT BOOKS

• For Homeschooling Parents: If you have already chosen a trusted core curriculum—such as Singapore Math (Primary Mathematics or Dimensions), Saxon Math, RightStart, Math-U-See, Math Mammoth, or Beast Academy / Art of Problem Solving (AoPS)—you do NOT need to switch books or risk curriculum gaps. Because our 12 Concept Domains follow the exact Concrete → Pictorial → Abstract progression of classical and mastery math, you can open to any topic for a 15-minute Kitchen-Table Lab whenever your child hits a new or challenging concept.
• For Classroom Teachers: If your school mandates Common Core (CCSS), Texas TEKS, Florida B.E.S.T., or Virginia SOL with Eureka Math, Illustrative Math, enVision, Big Ideas Math, or Bridges, use the Crosswalk on Pages 1–3 to plug any Concept Domain directly into your existing pacing calendar.

2. FOUR OPEN-AND-GO WAYS TO USE THIS PROGRAM AT THE KITCHEN TABLE OR IN CLASS

Way A: The 10-Minute Kitchen-Table or Whiteboard Launch (Before New Lessons)

Before introducing a new chapter in your regular math book (like regrouping, common denominators, or prime factoring), spend 10 minutes building the Domain's physical block model together and reading the 'What to Say & Ask' prompt. Once your student sees WHY the math works, their regular textbook lesson goes twice as fast.

Way B: Independent Self-Checking on the Stack (Worked Answers Here in Book 1 & Online)

To prevent students from peeking at answers in the back of their workbooks, printed answer tables are omitted from the student workbooks! Students solve on paper first and verify on their SC-6/SC-32 stack—while teachers have worked answers right on Pages A & B of every Domain in this guide and full worksheet tables at stackcalc.io/curriculum/answers.

Way C: The 'No-Tears' Hands-On Rescue When a Student Gets Stuck

When a child hits a wall with abstract rules (such as 'invert and multiply' or negative signs), pause the worksheet and open to the Domain's Target Misconceptions & Hands-On Block Remediations box. Ten minutes with Fraction Towers or Prime Factor Blocks turns frustration into an 'Aha!' moment.

Way D: 5-Day Pacing & 3-Tier Differentiation Across Worksheets Practice A through F

Each 11-page unit in Books 2A (K–2), 2B (3–5) & 4 (9–12) provides 6 leveled practice sheets for easy differentiation: Tier 1 Foundation & Intervention (Practice A–B), Tier 2 Grade-Level Core & Error Analysis (Practice C–D), and Tier 3 Challenge & Fast Finishers (Practice E–F & STEM Lab).

3. COMPLETE 6-VOLUME STACKCALC CURRICULUM LINEUP (100% SELF-CONTAINED BY GRADE BAND)

Book 1 (Educator) • Teaching with the Stack:	Master K–12 Teacher & Homeschool Guide, Block Games & Coaching Notes (46 pages).
Book 2A (Grades K–2) • Early Elementary Lab:	SC-6 K–2 Workbook (10 eleven-page units, Practice A–F, Quizzes, STEM & Cumulative Exams, 840 problems, 128 pp.).
Book 2B (Grades 3–5) • Upper Elementary Lab:	SC-6 3–5 Workbook (10 eleven-page units, Practice A–F, Quizzes, STEM & Cumulative Exams, 840 problems, 128 pp.).
Book 3 (Grades 6–8) • Middle School Math Lab:	SC-6 Middle School Workbook (10 units, 45 sections, 900 problems + 10 Unit Quizzes, 128 pp.).
Book 4 (Grades 9–12) • High School & STEM Lab:	SC-32 Scientific Workbook (10 eleven-page STEM units, Practice A–F, Quizzes & Cumulative Exams, 840 problems, 128 pp.).
Book 5 (Full-Year) • Stack Speed & Fluency:	Full-Year Speed Edition (15 Speed Lessons & 180 Unique Tests A1–D3 = 3,690 problems, 128 pp.).

Why There Is No [=] Key: You Already Teach This Way

If you grew up with a standard calculator—or worry about calculators in math—read this 2-minute page first.

IF YOU ARE SKEPTICAL OF CALCULATORS IN MATH, YOU ARE IN GOOD COMPANY!

Thoughtful parents and teachers often hesitate when they first see a calculator with an [ENTER] key and no [=] button: 'Won't a calculator make my child lazy at mental math? And why is the plus sign at the end?' Here is the reassuring truth: stack-based math (RPN) is not a button-pushing trick. It is the exact order in which you already teach math with physical blocks and vertical paper arithmetic at your kitchen table or whiteboard every day.

1. THE 'NOUN, NOUN, VERB' SECRET: PHYSICAL BLOCKS & PAPER ARE ALREADY RPN

How You Teach with Physical Blocks

1. 'Put 8 blocks on the tray.'
2. 'Now bring 5 more blocks.'
3. 'Now push them together (+)!'

You can't push blocks together before the second group is on the table!

How You Teach Vertical Paper Math

1. Write 8 on the top line (Line Y).
2. Write 5 on the line below (Line X).
3. Draw the line and add (+ → 13).

Both numbers exist on paper before the operation happens!

How the 4-Line Stack Verifies Paper Work

1. Type [8] [ENTER] → 8 sits on Line Y.
2. Type [5] → 5 sits on Line X.
3. Press [+] → adds $Y + X = 13$!

Line Y is the Waiting Shelf; Line X is the Workbench. Zero hidden state!

2. WHY THE STACK DEMANDS MATHEMATICAL UNDERSTANDING (INSTEAD OF BLIND COPYING)

STANDARD [=] CALCULATOR ON (8 + 4) / (9 - 3):

• The Crutch Trap: A student types 8 + 4 / 9 - 3 = left-to-right and gets 5.444... (WRONG!) because the calculator silently divided 4 / 9 first.
• Or Mindless Copying: Students copy ((8+4)/(9-3)) without knowing which operation happens first.
• Hidden Mistakes: A 1-line screen erases each number as soon as + is pressed, hiding where a student's thinking went wrong.

4-LINE STACK CALCULATOR ON (8 + 4) / (9 - 3):

• True Inside-Out Structure: With no parentheses keys, the student MUST understand order of operations first!
• Step 1: Solve 8 + 4 = 12 → 12 waits visibly on Line Y.
• Step 2: Solve 9 - 3 = 6 → 6 sits visibly on Line X.
• Step 3: Both 12 and 6 are visible together! Press [/] → 12 / 6 = 2. It mirrors paper-and-pencil mastery step for step!

3. ANSWERING THE THREE BIGGEST PARENT & TEACHER WORRIES

Worry 1: 'Won't a calculator weaken my child's mental math and paper-and-pencil discipline?'

Answer: Never when used the StackCalc way! Students solve every problem with blocks and paper FIRST, and never write out button lists. Because the stack has no [=] or parentheses keys, checking their answer reinforces inside-out order of operations.

Worry 2: 'We want less screen time, not more technology—and what about standard standardized tests?'

Answer: The SC-6/SC-32 is a distraction-free physical tool like an abacus (no Wi-Fi, games, or apps). And because students learn to decompose expressions from the inside out, they perform dramatically better on both non-calculator and standard-calculator tests!

Worry 3: 'I'm not a math specialist—how much prep does it take to teach this at home or in class?'

Answer: Zero prep! Every lesson is 100% self-contained with a word-for-word 'What to Say & Ask' script. Turn to Page vii and try the 4 three-minute exercises right now!

The 3-Minute Teacher Hands-On RPN Primer

Three golden rules, four 45-second try-it exercises, and a word-for-word Day 1 script to read to your class.

THE 3 GOLDEN RULES OF THE STACK (EVERYTHING YOU NEED TO KNOW)

1. [ENTER] = 'Put It on the Waiting Shelf': Press [ENTER] ONLY to separate two numbers you type back-to-back.

2. Action Keys Work Immediately: Press [+], [-], [x], or [/] AFTER both numbers are in place—never press [ENTER] after an operator!

3. Automatic Stack Lift: When you type a new number after an operation, your previous answer automatically lifts to Line Y!

TRY THESE 4 EXERCISES ON YOUR CALCULATOR RIGHT NOW (3 MINUTES TOTAL)

Try-It 1 (30 sec): Basic Addition — 8 + 5 = 13

Type [8] [ENTER] to lift 8 onto Shelf Y. Type [5] on Workbench X. Press [+] → 8 + 5 = 13!

Press Keys: 8 ENTER 5 +

Try-It 2 (45 sec): Two-Group PEMDAS Without Parentheses — (8 + 4) × (9 - 3) = 72

First group: [8] [ENTER] [4] [+] gives 12. Just start typing [9] [ENTER] [3] [-]! 12 lifts to Y while 6 forms on X. Press [x] → 72!

Press Keys: 8 ENTER 4 + 9 ENTER 3 - ×

Try-It 3 (45 sec): Two-Dot Fractions on the SC-6 — 1/3 + 1/4 = 7/12

Pressing [.] twice enters a fraction ([.] [1] [.] [3] = 1/3). Press [ENTER], type 1/4, and press [+] → displays 7/12 as a real fraction!

Press Keys: • 1 • 3 ENTER • 1 • 4 +

Try-It 4 (45 sec): Fixing Reversed Numbers Without Starting Over — 20 ÷ 4 = 5

Typed 4 then 20 when you meant 20 / 4? Never clear and start over! Press the Swap key to flip Y and X (Y: 20, X: 4), then press [/] → 5!

Press Keys: 4 ENTER 2 0 ↑↓ ÷

YOUR 60-SECOND DAY 1 SCRIPT, STACK HAND-SIGNALS & PARTNER PROTOCOLS

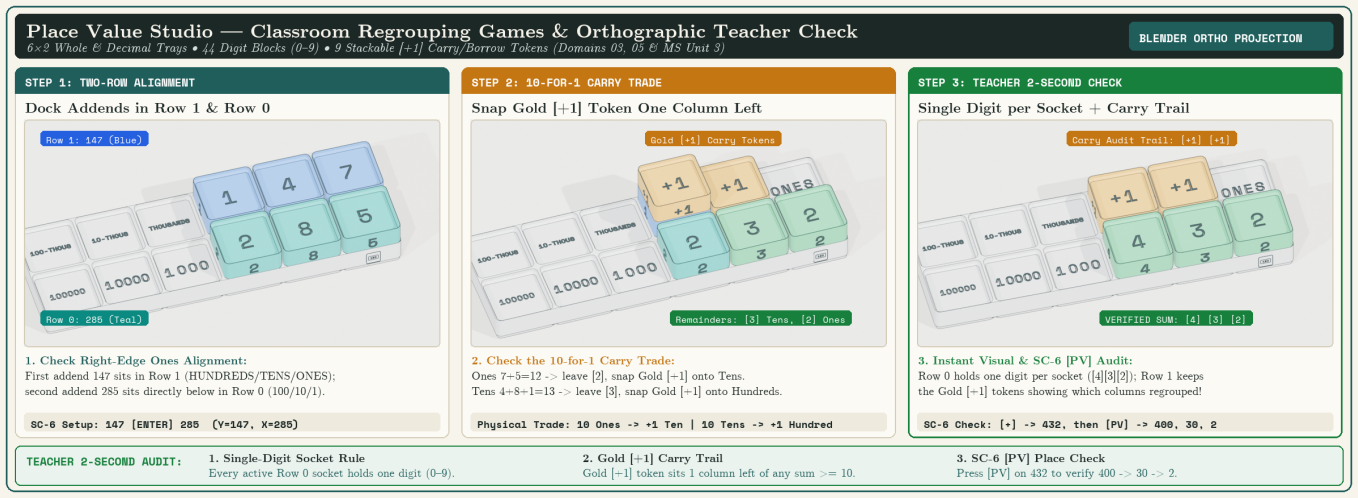
"Class, look at your calculator—there is no equals sign and no parentheses! It works just like your hands on your desk. Bottom Line X is your Workbench where you type a number; Line Y above it is your Waiting Shelf. To add 8 + 5, type 8 and press [ENTER] to lift 8 to the Waiting Shelf, type 5 on the Workbench, and press [+]!"

• 4 Stack Hand-Signals: Hands flat on desk = *X* (Workbench) • Chest height = *Y* (Waiting Shelf) • Shoulder height = *Z* • Overhead = *T*.

• Partner Roles & Two-Token Rule: Partner A (The Architect) builds the block model first (+10 token); Partner B (The Stack Navigator) verifies on the keypad (+10 token). Swap roles every 3 problems! End class with the 60-Second Gridfinity Socket Count-In (84 mm flush check).

Place Value Studio: Games & Visual Teacher Check

Classroom games and 2-second orthographic visual checks for the 6×2 Whole/Decimal Trays, Digit Blocks (0–9), and [+1] Carry Tokens.



CLASSROOM GAMES & HANDS-ON ACTIVITIES (TIED TO CONCEPT DOMAINS & WORKBOOKS)

Game 1: 'Race to 1,000 Carry Relay' (Domains 03 & 05 • Grades 2–5 • 10-Min Partner Relay)

Partners start at 000 on the 6×2 Whole Number Tray. Take turns rolling two dice (or spinning 0–9) to form a 2-digit or 3-digit addend in Row 0 (100/10/1) and add it to the running total in Row 1. Whenever a column sum reaches 10+, the player calls 'Regroup!', leaves the ones digit in Row 0, and snaps a Gold [+1] Carry Token one column left. First pair to click [1] into THOUSANDS (1000) and verify on the SC-6 wins!

Check Keys (e.g., 147 + 285 = 432): `1` `4` `7` `ENTER` `2` `8` `5` `+` `123`

Game 2: 'The Decimal Point Shifter (×10 & ÷10 Slide)' (Domain 05 & MS Unit 03 • Grades 4–6 • 10 Min)

Dock the Whole Number Tray (100,000 to 1) and Decimal Tray (0.1 to 0.000001) side-by-side at the raised tactile decimal point (•). Call out a starting decimal (e.g., 4.75) and a power-of-10 action (× 10, × 100, or ÷ 10). Students physically slide every digit block one or two columns left (for ×10/×100) or right (for ÷10) across the stationary decimal point—seeing why digits shift columns rather than the point moving!

Check Keys (4.75 × 10 = 47.5): `4` `•` `7` `5` `ENTER` `1` `0` `×` `123`

Game 3: 'The Borrowing Bank Vault (Zero-Placeholder Heist)' (Domain 05 • Grades 3–6 • 12 Min)

Set a 'Vault Target' containing zeros in Row 1 (e.g., 502) and a subtraction code in Row 0 (e.g., 168). Because Ones (2 - 8) and Tens (0 - 6) cannot subtract directly, the 'Banker' partner must un-group 1 Hundred ([5] → [4]), trade across the zero Tens ([0] → [9]), and snap the Gold [+1] regrouping token onto the Ones column (12 - 8 = 4) to unlock 334.

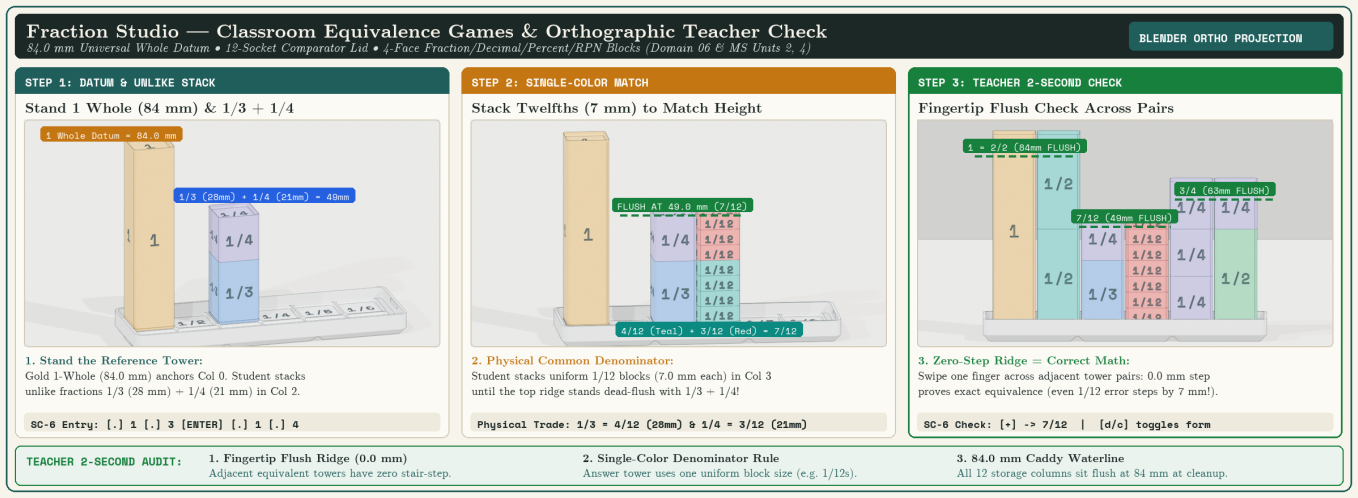
Check Keys (502 − 168 = 334): `5` `0` `2` `ENTER` `1` `6` `8` `-` `123`

HOW TO CHECK STUDENTS ARE DOING IT RIGHT (2-SECOND WALK-BY AUDIT)

- Check 1 — Single-Digit Socket Rule: Glance at Row 0—every active column must hold exactly ONE digit block (0–9). If a student stacked two digit blocks without trading, pause and ask: 'Did this column reach 10?'
- Check 2 — Gold [+1] Carry Trail: Every Gold [+1] token in Row 1 must sit one column to the LEFT of the column that crossed 10.
- Check 3 — SC-6 [PV] Place Decomposition: Have the student press [PV] on their SC-6 answer (432 → 400 → 30 → 2) while pointing to each tray column from left to right.

Fraction Studio: Games & Visual Teacher Check

Classroom games and 2-second orthographic flush checks for the 84.0 mm Proportional Fraction Towers & 12-Socket Comparator Lid.



CLASSROOM GAMES & HANDS-ON ACTIVITIES (TIED TO CONCEPT DOMAINS & WORKBOOKS)

Game 1: 'The 84 mm Flush Tower Challenge (Make 1 Whole)' (Domain 06 • Grades 3–5 • 10 Min)

Stand the Gold 1-Whole block (84.0 mm) in Col 0 of the 12-Socket Comparator Lid. Call out a constraint—e.g., 'Build 1 Whole in Col 1 using exactly 4 blocks from at least 3 different denominator colors!' (e.g., 1/2 + 1/4 + 1/6 + 1/12 = 42 + 21 + 14 + 7 = 84.0 mm). Students swipe a finger across the top ridge to prove 0.0 mm step, then verify with two-dot fraction entry on the SC-6.

Check Keys (1/2 + 1/4 + 1/6 + 1/12 = 1): [•] [1] [•] [2] [ENTER] [•] [1] [•] [4] [+] [•] [1] [•] [6] [+] [•] [1] [•] [1] [2] [+]

Game 2: 'Two-Dot Common Denominator Duel (1/3 + 1/4 = 7/12)' (Domain 06 & MS Unit 02 • Grades 4–6)

Partner A stacks two unlike fractions in Col 2 (e.g., 1/3 [28 mm] + 1/4 [21 mm] = 49.0 mm). Partner B must find a SINGLE-COLOR block family (twelfths, 7.0 mm each) and stack enough identical blocks in Col 3 (seven 1/12 blocks = 49.0 mm) to stand 100% flush with Partner A's tower. Both partners key the sum on the SC-6 and press [d/c] to inspect proper/improper forms.

Check Keys (1/3 + 1/4 = 7/12): [•] [1] [•] [3] [ENTER] [•] [1] [•] [4] [+] [f] [d/c]

Game 3: 'Mystery Tower Blind Fingertip Audit & 4-Face Spin' (Domain 06 & MS Unit 04 • Grades 3–6)

Student A builds a target fraction tower (e.g., three 1/4 blocks = 3/4 = 63.0 mm) and challenges Student B to build an equal-height tower using NO fourths (e.g., 1/2 + 1/4, or four 1/6 + one 1/12 = 63.0 mm). Student A closes their eyes and runs one fingertip across both tops to judge if they are flush, then both rotate a block 90° at a time to read Fraction (1/4), Decimal (0.25), Percent (25%), and RPN Key (.1.4).

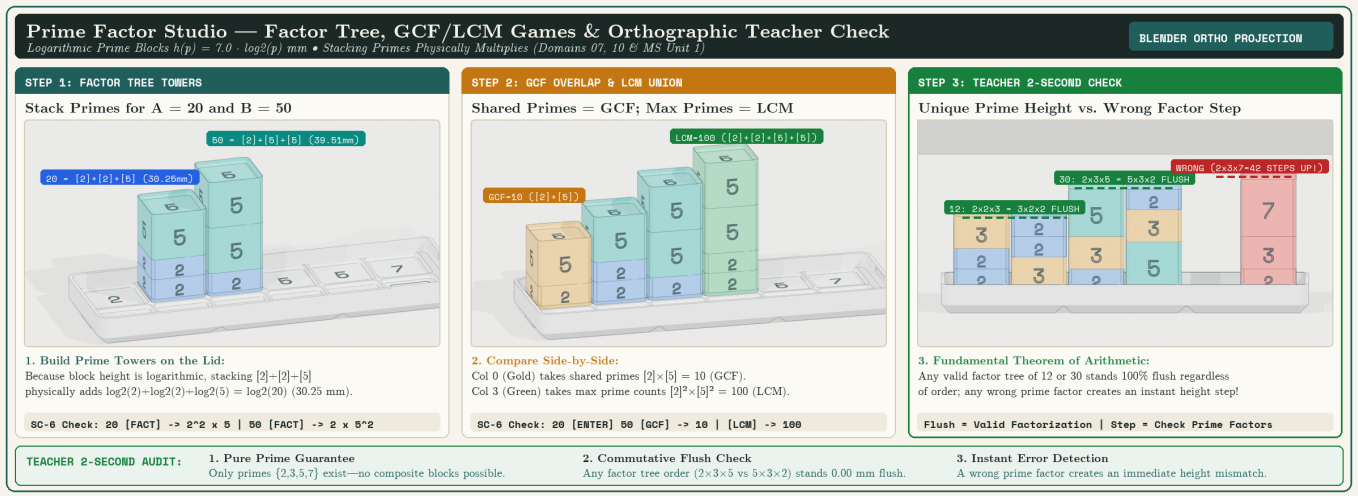
Check Keys (4/6 + 1/12 = 3/4): [•] [4] [•] [6] [ENTER] [•] [1] [•] [1] [2] [+]

HOW TO CHECK STUDENTS ARE DOING IT RIGHT (2-SECOND WALK-BY AUDIT)

- Check 1 — The 0.0 mm Fingertip Flush Test: Even the smallest fraction error (1/12) creates a bold 7.0 mm physical stair-step! If two adjacent towers are dead-flush across the top ridge, the student's equivalence or common denominator is 100% mathematically correct.
- Check 2 — Single-Color Answer Tower: When adding unlike fractions (1/3 + 1/4), check that the student's answer tower in the adjacent socket uses ONE uniform block color (seven 1/12 blocks) rather than mixed denominators.
- Check 3 — 3-Second Cleanup Waterline: When packed into the 12-column caddy, every column sits dead-flush at 84.0 mm—any missing block leaves an obvious gap.

Prime Factor Studio: Games & Visual Teacher Check

Classroom games and 2-second orthographic checks for the Logarithmic Prime Factor Blocks ($h(p) = 7.0 \cdot \log_2(p)$ mm) & 12-Socket Comparator Lid.



CLASSROOM GAMES & HANDS-ON ACTIVITIES (TIED TO CONCEPT DOMAINS & WORKBOOKS)

Game 1: 'The Commutative Skyscraper Race (2 × 3 × 5 = 30)' (Domains 07 & 10 • Grades 3–6 • 10 Min)

Call out a composite number (e.g., 12, 30, or 42). Partner A factors it using one factor tree (e.g., $30 = 6 \times 5 \rightarrow [2] + [3] + [5]$) and stacks the prime blocks in Col 2. Partner B uses a different factor tree ($30 = 10 \times 3 \rightarrow [5] + [2] + [3]$) in Col 3. Because block height follows $h(p) = 7.0 \cdot \log_2(p)$ mm ($[2] = 7.0$ mm $= 1\,U$, $[3] = 11.09$ mm, $[5] = 16.25$ mm), both towers stand 100% flush at 34.35 mm—proving commutativity and unique factorization physically!

Check Keys (Factor 30 → 2 × 3 × 5): 3 0 ENTER g 人

Game 2: 'Impossible Impostor (Fundamental Theorem Challenge)' (Domain 10 & MS Unit 01 • Grades 5–8)

Student A builds a prime tower (e.g., $12 = [2] + [2] + [3] = 25.10$ mm) and challenges Student B to match its exact height using DIFFERENT prime blocks (e.g., trying $[5] + [2] = 10 = 23.25$ mm or $[2] + [7] = 14 = 26.65$ mm). Students quickly discover that NO other combination of primes will ever stand flush—experiencing the Fundamental Theorem of Arithmetic in their hands!

Check Keys (Compare 12 vs 14): 1 2 g 人 1 4 g 人

Game 3: 'Side-by-Side GCF Overlap & LCM Union Duel' (Domain 10 & MS Unit 01 • Grades 5–8 • 12 Min)

Given $A = 20$ and $B = 50$, students stack Tower A ([2]+[2]+[5]) in Col 1 and Tower B ([2]+[5]+[5]) in Col 2. To build GCF(20,50) = 10 in Col 0, they copy only the SHARED prime blocks present in both towers ([2]+[5]). To build LCM(20,50) = 100 in Col 3, they stack the tallest count of each prime ([2]+[2]+[5]+[5]), then verify [FACT], [GCF], and [LCM] on the SC-6!

Check Keys (GCF & LCM of 20 and 50): 2 0 ENTER 5 0 GCF 2 0 ENTER 5 0 f LCM

HOW TO CHECK STUDENTS ARE DOING IT RIGHT (2-SECOND WALK-BY AUDIT)

- Check 1 — Pure Prime Guarantee: Because the kit contains ONLY prime blocks ({2, 3, 5, 7}), students can never stop a factor tree early at a composite number like 4, 6, 8, or 9.
- Check 2 — Logarithmic Flush Check: Any two valid factor trees for the same number stand dead-flush (0.000 mm step), while any wrong prime factor creates an immediate height step (see red tower in Step 3 above).
- Check 3 — GCF + LCM Height Symmetry: On the comparator lid, GCF is always the shortest tower (left), LCM is the tallest tower (right), and h(GCF) + h(LCM) = $h(A)$ + h(B).

Master K–12 Homeschool & Classroom Crosswalk

Locate your homeschool program (Singapore, Saxon, RightStart, AoPS) or school textbook/standard to find the matching Concept Domain.

Concept Domain	Standards (CCSS•TEKS•BEST)	Homeschool (Singapore / Saxon / AoPS)	School (Eureka / Big Ideas / Bridges)	Guide & Lab
Dom 01: Counting & Step	K.CC.B.4 • TEKS K.2A • MA.K.NSO.1	**Singapore EB/PK • RightStart A • Saxon K**	Eureka GK M1 • Bridges GK U1	**pp. 4–5 • 2A U01,U10**
Dom 02: Part-Part-Whole	1.OA.B.4 • TEKS 1.3B • MA.1.AR.1	**Singapore 1A Ch2 • RightStart B • Math-U-See α**	Eureka G1 M1 • Bridges G1 U2	**pp. 6–7 • 2A U02,U06**
Dom 03: Make-10 & Carry	1.NBT.B.2 • TEKS 1.3D • MA.1.NSO.2	**Singapore 1A Ch8 • Saxon 1–2 • RightStart B**	Eureka G1 M2 • Big Ideas G1 Ch3	**pp. 8–9 • 2A U03,07,08**
Dom 04: Doubles & Groups	2.OA.C.4 • TEKS 2.4A • MA.2.AR.3	**Singapore 2A Ch4 • Saxon 2–3 • Math-U-See β**	Eureka G2 M6 • Bridges G2 U1	**pp. 10–11 • 2A U04,09; 2B U01**
Dom 05: Multi-Digit Stack	4.OA.A.3 • TEKS 4.4H • MA.4.AR.1	**Singapore 3A/4A • Saxon 5/4 • Beast Acad 3–4**	Eureka G4 M3 • Big Ideas G4 Ch2	**pp. 12–13 • 2A U05; 2B U02,04,08**
Dom 06: Fraction Towers	5.NF.A.1 • TEKS 5.3H • MA.5.FR.2	**Singapore 4A/5A • Saxon 6/5 • Math-U-See ε**	Eureka G5 M3 • Bridges G5 U2	**pp. 14–15 • 2B U03,06,09,10**
Dom 07: Prime Factors & Trees	6.NS.B.4 • TEKS 6.7A • MA.6.NSO.3	**Singapore 5A/6A • Saxon 7/6 • AoPS Prealgebra**	Illustrative G6 U4 • Big Ideas G6 Ch1	**pp. 16–17,29 • 2B U05,07; 4 U01**
Dom 08: Quadratics & Poly	HSA.APR • TEKS A.10E • MA.912.AR.3	**Singapore 8A Ch5 • Saxon Alg 1 • AoPS Intro Alg**	Illustrative Alg1 U6 • Big Ideas Alg1	**pp. 18–19 • Book 4 U02–U04**
Dom 09: Trig & 2D Vectors	HSG.SRT.8 • TEKS G.9A • MA.912.T.1	**Singapore Add Math • Saxon Geom/Adv • AoPS Geom**	Illustrative Geom U4 • Big Ideas Geom	**pp. 20–21 • Book 4 U07–U08**
Dom 10: Exponents & Logs	HSF.LE.4 • TEKS 2A.5D • MA.912.AR.5	**Saxon Adv Math • Foerster Alg 2 • AoPS Int Alg**	Illustrative Alg2 U4 • Big Ideas Alg2	**pp. 22–23 • Book 4 U05–U06**
Dom 11: Stats & Combinatorics	6.SP / HSS.IC • TEKS S.4B • MA.912.DP	**AoPS Counting & Prob • Singapore 7B/Add Math**	Big Ideas G6 Ch9–10 & AP Statistics	**pp. 24–25,30 • Book 4 U09**
Dom 12: Binary & Hex Logic	AP CSP Big Idea 2 • TEKS CS • MP5	**AoPS Intro Number Theory (Base-N) • CS Logic**	AP Computer Science Principles / CS A	**pp. 26–27 • Book 4 U10**

How Homeschooling Parents & Classroom Teachers Use This Crosswalk

Keep your current math book! Whenever your student reaches one of the 12 core topics above in Singapore, Saxon, RightStart, AoPS, or your school textbook, open to the listed Guide & Lab pages for a 15-minute hands-on block & paper-first investigation.

Elementary Homeschool & Classroom Concordance (Grades K–5)

Lesson-by-lesson mapping across Singapore Math (Dimensions/Primary), Bridges in Mathematics, Common Core (CCSS), and Eureka/enVision.

Grade	Curriculum / Standard	Mathematical Concept Topic	Manipulative / Model	Teacher Guide	Student Lab
Kindergarten	K.CC.B.4 • Dimensions KA, Ch 1, Lesson 1.2	**One-to-One Counting & Step Tokens**	Place Value Studio (Digit Blocks & [+1] Tokens)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.CC.A.3 • Dimensions KA, Ch 2, Lesson 2.3	**Zero and Empty Sets**	Place Value Studio ([0] Digit Block)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.OA.A.3 • Dimensions KA, Ch 4, Lesson 4.1	**Number Bonds of 5**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Kindergarten	K.OA.A.1 • Dimensions KB, Ch 7, Lesson 7.2	**Addition within 10 on the Stack**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Kindergarten	K.OA.A.2 • Dimensions KB, Ch 8, Lesson 8.1	**Subtraction as Missing Part**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Grade 1	1.NBT.A.1 • Dimensions 1A, Ch 1, Lesson 1.3	**Counting & Comparing Numbers to 10**	Place Value Studio (Digit Blocks & [+1] Tokens)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Grade 1	1.OA.C.6 • Dimensions 1A, Ch 2, Lesson 2.1	**Making Number Bonds to 6 and 8**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Grade 1	1.OA.C.6 • Dimensions 1A, Ch 2, Lesson 2.2	**Making Number Bonds to 10**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Grade 1	1.OA.B.4 • Dimensions 1A, Ch 2, Lesson 2.4	**Decomposing Numbers & Missing Parts**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Grade 1	1.OA.C.6 • Dimensions 1A, Ch 3, Lesson 3.1	**Addition as Part-Part-Whole**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Grade 1	1.OA.C.6 • Dimensions 1A, Ch 4, Lesson 4.2	**Subtraction as Comparing Sets**	Place Value Studio (Two-Row Tray)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Grade 1	1.NBT.B.2 • Dimensions 1A, Ch 7, Lesson 7.1	**Place Value Tens & Ones (11-20)**	Place Value Studio (Digit Blocks & [+1] Carry)	**Dom 03 (pp.8–9)**	**Lab Unit 03**
Kindergarten	K.CC.B.4 • Unit 1, Mod 1, Session 1	**Welcome to Math: Counting Objects**	Place Value Studio (Digit Blocks & [+1] Tokens)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.CC.B.4 • Unit 1, Mod 1, Session 2	**Counting & Five-Benchmarks**	Place Value Studio (Digit Blocks & [+1] Tokens)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.CC.B.5 • Unit 1, Mod 2, Session 1	**Place Value Tray & Counting On**	Place Value Studio (Digit Blocks & [+1] Tokens)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.CC.A.3 • Unit 1, Mod 2, Session 3	**Zero & The [0] Digit Block**	Place Value Studio ([0] Digit Block)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.OA.A.1 • Unit 1, Mod 3, Session 2	**The Step Counter: Adding One More**	Place Value Studio (Digit Blocks & [+1] Tokens)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.CC.C.6 • Unit 2, Mod 1, Session 2	**Comparing Sets: Which Has More?**	Place Value Studio (Two-Row Tray)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.OA.A.3 • Unit 2, Mod 2, Session 3	**Number Bonds to 5**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Kindergarten	K.OA.A.2 • Unit 4, Mod 2, Session 1	**Part-Part-Whole Addition**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Grade 1	1.OA.C.6 • Unit 1, Mod 2, Session 3	**Place Value & Complements to 10**	Place Value Studio (Digit Blocks & [+1] Carry)	**Dom 03 (pp.8–9)**	**Lab Unit 03**
Grade 1	1.OA.C.6 • Unit 2, Mod 1, Session 2	**Combinations to Ten**	Place Value Studio (Digit Blocks & [+1] Carry)	**Dom 03 (pp.8–9)**	**Lab Unit 03**
Grade 1	1.OA.B.4 • Unit 2, Mod 1, Session 4	**Finding the Missing Addend**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Grade 1	1.OA.C.6 • Unit 2, Mod 3, Session 3	**Making 10 Addition Strategies**	Place Value Studio (Digit Blocks & [+1] Carry)	**Dom 03 (pp.8–9)**	**Lab Unit 03**

Plugging Into Your Homeschool Rhythm or Classroom Pacing Guide

Match by your curriculum chapter (Singapore Dimensions, Bridges, Big Ideas) or standard code (e.g., 1.OA.C.6, 5.NF.A.1, 6.EE.A.2) above, then turn to the listed Concept Domain in Part I (or Part II on pp. 28–32 for Middle School Grades 6–8).

Middle School, High School & Homeschool Concordance (Grades 6–12)

Section-by-section mapping across Singapore Dimensions 6–8, Saxon/AoPS, Big Ideas Math, Illustrative Math, and AP Math/CS.

Grade	Curriculum / Standard	Mathematical Concept Topic	Manipulative / Model	Teacher Guide	Student Lab
Grade 6	6.NS.C.5 • Dimensions 6A, Ch 1, Lesson 1.2	**Negative Integers on Stack**	Place Value Studio & 4-Register Stack	**Dom 05 (pp.12–13)**	**Lab Unit 05**
Grade 6	6.EE.A.2 • Dimensions 6A, Ch 3, Lesson 3.1	**Syntax Trees & Algebraic Expressions**	Prime Factor Studio & 4-Register Stack	**Dom 07 (pp.16–17)**	**Lab Unit 07**
Grade 6	6.EE.A.2 • Dimensions 6A, Ch 3, Lesson 3.3	**Postorder Traversal Evaluation**	Prime Factor Studio & 4-Register Stack	**Dom 07 (pp.16–17)**	**Lab Unit 07**
Grade 6	6.NS.C.8 • Dimensions 6B, Ch 7, Lesson 7.2	**Distance in the Coordinate Plane**	Place Value Studio (Two-Row Tray)	**Dom 08 (pp.18–19)**	**Lab Unit 08**
Grade 7	7.EE.A.1 • Dimensions 7A, Ch 3, Lesson 3.2	**Linear Expressions & Register Depth**	Prime Factor Studio & 4-Register Stack	**Dom 07 (pp.16–17)**	**Lab Unit 07**
Grade 7	7.EE.A.1 • Dimensions 7A, Ch 3, Lesson 3.4	**Binomial Expansion & Factoring**	Place Value & Prime Factor Studios	**Dom 08 (pp.18–19)**	**Lab Unit 08**
Grade 7	7.SP.C.8 • Dimensions 7B, Ch 11, Lesson 11.2	**Sample Space & Probability Ratios**	Fraction Studio (84 mm Ratio Towers)	**Dom 11 (pp.24–25)**	**Lab Unit 11**
Grade 8	HSA.APR.A.1 • Dimensions 8A, Ch 5, Lesson 5.2	**Factoring Quadratics via Prime Pairs**	Place Value & Prime Factor Studios	**Dom 08 (pp.18–19)**	**Lab Unit 08**
Grade 8	HSA.REI.B.4 • Dimensions 8A, Ch 5, Lesson 5.4	**Difference of Two Squares**	Place Value & Prime Factor Studios	**Dom 08 (pp.18–19)**	**Lab Unit 08**
Grade 8	8.G.B.7 • Dimensions 8A, Ch 7, Lesson 7.3	**The Pythagorean Theorem**	Place Value Studio & 4-Register Stack	**Dom 08 (pp.18–19)**	**Lab Unit 08**
Grade 6	6.EE.A.2 • Chapter 1, Section 1.3	**Order of Operations & Syntax Trees**	Prime Factor Studio & 4-Register Stack	**Dom 07 (pp.16–17)**	**Lab Unit 07**
Grade 6	6.EE.A.2 • Chapter 3, Section 3.2	**Writing Algebraic Expressions as Trees**	Prime Factor Studio & 4-Register Stack	**Dom 07 (pp.16–17)**	**Lab Unit 07**
Grade 6	6.EE.A.2 • Chapter 3, Section 3.3	**Postorder Traversal Evaluation**	Prime Factor Studio & 4-Register Stack	**Dom 07 (pp.16–17)**	**Lab Unit 07**
Grade 6	6.NS.C.8 • Chapter 6, Section 6.5	**Distance in the Coordinate Plane**	Place Value Studio (Two-Row Tray)	**Dom 08 (pp.18–19)**	**Lab Unit 08**
Grade 7	7.NS.A.1 • Chapter 1, Section 1.2	**Adding & Subtracting Rational Numbers**	Place Value Studio & 4-Register Stack	**Dom 05 (pp.12–13)**	**Lab Unit 05**
Grade 7	7.EE.A.1 • Chapter 3, Section 3.2	**Simplifying Multi-Operator Expressions**	Prime Factor Studio & 4-Register Stack	**Dom 07 (pp.16–17)**	**Lab Unit 07**
Grade 8	8.EE.B.5 • Chapter 4, Section 4.2	**Slope of a Line (Rise / Run)**	Coordinate Slope Triangle & 4-Register Stack	**Dom 08 (pp.18–19)**	**Lab Unit 08**
Grade 8	8.G.B.7 • Chapter 7, Section 7.3	**The Pythagorean Theorem**	Right-Triangle Diagram & 4-Register Stack	**Dom 08 (pp.18–19)**	**Lab Unit 08**
Grade 8	8.G.B.8 • Chapter 7, Section 7.5	**The Distance Formula Pipeline**	Right-Triangle Diagram & 4-Register Stack	**Dom 08 (pp.18–19)**	**Lab Unit 08**
High School	HSG.SRT.C.8 • Geometry, Ch 9, Section 9.1	**Pythagorean Theorem & Run/Rise**	Right-Triangle Diagram & SC-32	**Dom 08 (pp.18–19)**	**Lab Unit 08**
High School	HSG.SRT.C.8 • Geometry, Ch 9, Section 9.4	**Tangent, Sine & Cosine Ratios**	Right-Triangle Ratio Diagram & SC-32	**Dom 09 (pp.20–21)**	**Lab Unit 09**
High School	HSN.VM.A.1 • Geometry, Ch 9, Section 9.6	**Vectors in the Coordinate Plane**	2D Vector Right-Triangle Diagram	**Dom 09 (pp.20–21)**	**Lab Unit 09**
High School	HSN.VM.A.1 • Geometry, Ch 9, Section 9.7	**Polar Coordinates & Conversion**	2D Vector Right-Triangle Diagram	**Dom 09 (pp.20–21)**	**Lab Unit 09**
High School	HSF.LE.A.1 • Algebra 2, Ch 6, Section 6.1	**Exponential Growth Models**	Prime Factor Studio (Logarithmic Towers)	**Dom 10 (pp.22–23)**	**Lab Unit 10**

Plugging Into Your Homeschool Rhythm or Classroom Pacing Guide

Match by your curriculum chapter (Singapore Dimensions, Bridges, Big Ideas) or standard code (e.g., 1.OA.C.6, 5.NF.A.1, 6.EE.A.2) above, then turn to the listed Concept Domain in Part I (or Part II on pp. 28–32 for Middle School Grades 6–8).

PART I • DOMAIN 01 • COUNTING & CARDINALITY • PRE-K – KINDERGARTEN (SC-6 ELEMENTARY)

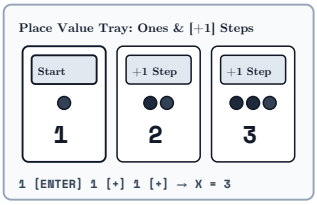
1-to-1 Correspondence: Stepping Up by One

Core Concept Lesson • Bridges in Mathematics (Numbers to 5 & 10) • Standard: CCSS.MATH.K.CC.B.4

1. MATHEMATICAL CONCEPT & STACK APPROACH: Establish one-to-one correspondence between physical digit blocks and step tokens on the Place Value Tray and verify counts.

Counting by ones is repeated addition of the unit 1 ($n \to n + 1$). On the Place Value Tray and the stack, your current count waits first while you bring in 1 more step, then both combine into the new total.

Classroom Hook: Close your eyes and clap slowly 4 times. Before opening your eyes, which Digit Block belongs in the Ones socket of your Place Value Tray and what number waits on Line X?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Digit Blocks 0–9 & Gold [+1] Step Tokens)

Hands-On Setup: Place the [1] Digit Block into the Ones socket of the Place Value Tray. Keep Gold [+1] step tokens ready beside the tray.

Predict First (Before Calculator): When we place 1 Gold [+1] step token and step the Ones block up by 1, what digit block replaces [1]? (Prediction: [2])

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Place 1 Gold [+1] step token on the tray first, then press [1] [+] to match the count you just touched."

2B. AVOID SAYING (RULE WITHOUT REASON):

"Just press 1 + 1 + 1 as fast as you can until the screen says 5."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [1] [ENTER] [1] [+] [1] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[1]	1	First count placed: 1 enters register X
[ENTER]	1	1 copied to register Y: holds the current count
[1]	1	Gold [+1] step token added: 1 enters register X
[+]	2	Press [+]: 1 + 1 = 2! Swap Ones block to [2]
[1]	1	Next Gold [+1] step token added: 1 enters X
[+]	3	Press [+]: 2 + 1 = 3! Tray [3] and screen match perfectly

4. CONTEXT (How Ten Fingers & an Empty Shell Created Place Value) • From Blocks to the Stack (SC-6 Manual) — Ch. 3: Counting, Subitizing & Place Value + Appendix B: Station F.1

Long before paper or keyboards existed, ancient shepherds counted flocks with clay pebbles ('calculi' in Latin—the root of the word 'calculate'!) and Maya astronomers drew a shell for zero so an empty place-value column had its own symbol. Stepping one [+1] token at a time on the Place Value Tray makes the 1-to-10 counting staircase tangible.

Core Investigation (Student Warm-Up A): Place the [4] Digit Block in the Ones column of your Place Value Tray and add 1 Gold [+1] step token. What total quantity do you have now?

Worked Answer: 5 ([5] Digit Block; 4 + 1 = 5). (All Keys: stackcalc.io/curriculum/answers#unit-01)

Verification Keys: [4] [ENTER] [1] [+]

More, Less & Zero: Finding Differences

Follow-Up Extension (Common Core / TEKS (Eureka, enVision, Bridges) • CCSS.MATH.K.CC.C.6) + Misconception Remediations & Lab Verification

1. FOLLOW-UP EXTENSION LESSON: CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Comparison subtraction finds the difference ('how many more?') between two quantities. Placing the larger group on the top row (Shelf Y) and the smaller group on the bottom row (Shelf X) shows how much remains when the matching part is taken away.

Blocks & Prediction: Place the [5] Digit Block in Row 1 (top row) and the [2] Digit Block in Row 0 (bottom row) of the Ones column on the Place Value Tray. Prediction: Look at the two rows on the tray. Which row has the larger quantity? How many more is 5 than 2?

Extension Key Sequence: [5] [ENTER] [2] [-]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[5]	5	Row 1 quantity (5) enters register X
[ENTER]	5	5 lifted to register Y (first set)
[2]	2	Row 0 quantity (2) enters register X
[-]	3	Minus subtracts X from Y: Row 1 has 3 more!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Reciting number names faster than placing physical step tokens (losing 1-to-1 correspondence).

Hands-On Block & Stack Fix: Have the student place 1 Gold [+1] token on the Place Value Tray and step the Ones Digit Block first, say the number aloud, and only then press [1] [+] so the tray and Register X stay locked in step.

Misconception 2: Confusing the action of adding 1 more with the total quantity on the tray (cardinality).

Hands-On Block & Stack Fix: Ask the student to point to the Digit Block in the Ones socket of the Place Value Tray and verify that its numeral matches Register X on the screen.

3. HANDS-ON LAB COACHING & STACK VERIFICATION KEYS (DOMAIN 01 WARM-UP B & INVESTIGATIONS 1–4)

Warm-Up B (Ext.): Row 1 holds [6] and Row 0 holds [2] on your Place Value Tray. Which row has more, and how many more does it have?
Check Keys: [6] [ENTER] [2] [-]
Ans: Row 1 has 4 more (6 - 2 = 4).

Investigation 1: Place [3] in the top row and [2] in the bottom row of your Place Value Tray. Combine both groups—what is the total?
Check Keys: [3] [ENTER] [2] [+]
Ans: 5

Investigation 2: You have 5 ones on the tray and take away 2. What Digit Block shows how many ones remain?
Check Keys: [5] [ENTER] [2] [-]
Ans: 3

Investigation 3: You have 3 blocks on the tray and take away all 3. How many blocks are left, and which Digit Block shows an empty tray?
Check Keys: [3] [ENTER] [3] [-]
Ans: 0 (Zero)

Investigation 4: Start with [1] in the Ones socket and add one more [+1] token three times in a row. What count do you reach?
Check Keys: [1] [ENTER] [1] [+] [1] [+] [1] [+]
Ans: 4

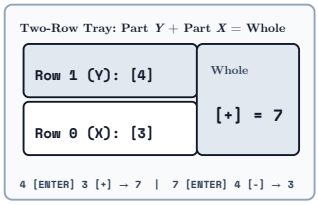
Composing Numbers: Part + Part = Whole

Core Concept Lesson • Common Core / TEKS (Eureka, enVision, Bridges) (Number Bonds & Part-Part-Whole) • Standard: CCSS.MATH.1.OA.C.6

1. MATHEMATICAL CONCEPT & STACK APPROACH: Physically compose two parts on the two rows of the Place Value Tray into a whole and connect to addition.

Every addition problem composes two independent parts into a single whole (Part 1 + Part 2 = Whole). Placing Part 1 on the top row (Shelf Y) and Part 2 on the bottom row (Shelf X) makes both parts visible before they are joined.

Classroom Hook: Place [7] as the Whole and cover Row 0 on your Place Value Tray so only Row 1 ([4]) shows. How does the stack uncover the hidden part?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Two-Row Tray & Digit Blocks 0–9)

Hands-On Setup: Place the [4] Digit Block in Row 1 (Part 1) and the [3] Digit Block in Row 0 (Part 2) of the Ones column on your Place Value Tray.

Predict First (Before Calculator): Combine both parts in the Ones column: 4 + 3 = 7 (swap to the [7] Digit Block).

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Row 1 ([4]) sits on Waiting Shelf Y and Row 0 ([3]) sits on Workbench X—press [+] to combine both parts into the Whole!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Move the 4 across the equals sign and change its sign to minus."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [4] [ENTER] [3] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[4]	4	Part 1 (4) enters register X
[ENTER]	4	4 copied up to register Y; represents Row 1 (Part 1)
[3]	3	Part 2 (3) enters register X; represents Row 0 (Part 2)
[+]	7	Plus combines Part 1 (Y) + Part 2 (X) = Whole (7) in X!

4. CONTEXT (Part-Part-Whole Number Bonds & Friends of Ten) • From Blocks to the Stack (SC-6 Manual) — Ch. 3: Number Bonds, Ch. 8: Missing Addend Mysteries & Appendix B: Station F.2

When students place two parts on the top and bottom rows of the Place Value Tray, they see addition and subtraction as two views of the exact same relationship: joining Part 1 and Part 2 builds the Whole, while removing Part 1 from the Whole reveals Part 2.

Core Investigation (Student Warm-Up A): Place [5] in Row 1 (Part 1) and [4] in Row 0 (Part 2) of your Place Value Tray. Combine the two parts to find the Whole.

Worked Answer: 9 ([9] Digit Block; 5 + 4 = 9). (All Keys: stackcalc.io/curriculum/answers#unit-02)

Verification Keys: [5] [ENTER] [4] [+]

Decomposing the Whole: Finding the Missing Part

Follow-Up Extension (Bridges in Mathematics • CCSS.MATH.1.OA.B.4) + Misconception Remediations & Lab Verification

1. FOLLOW-UP EXTENSION LESSON: CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Finding a missing addend (6 + ? = 10) is the inverse of composing a sum. By placing the known Whole (10) on the top shelf and subtracting the known Part (6), you isolate the unknown part.

Blocks & Prediction: Place [1][0] in Row 1 of the Place Value Tray (Whole = 10) and [6] in Row 0 (Known Part 1 = 6). Prediction: What missing part combines with 6 to make 10? (10 - 6 = 4).

Extension Key Sequence: [1] [0] [ENTER] [6] [-]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	Enter Whole (10)
[0]	10	10 in register X
[ENTER]	10	Whole (10) lifted to register Y
[6]	6	Known Part 1 (6) enters register X
[-]	4	Minus subtracts known Part from Whole: Missing Part is 4!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Pressing [+] immediately after the first number without pressing [ENTER] to separate Part 1 and Part 2.

Hands-On Block & Stack Fix: Point to the two rows of the Place Value Tray: pressing [ENTER] is the physical divider that lifts Row 1 (Part 1) up to Register Y before Row 0 (Part 2) is typed into Register X.

Misconception 2: Adding the whole and the known part when solving for a missing addend (e.g., computing 10 + 6 instead of 10 − 6).

Hands-On Block & Stack Fix: Place [1][0] in Row 1 (Whole) and [6] in Row 0 (Known Part) of the Place Value Tray; show that the Whole (10) goes into Register Y, the known part (6) enters X, and [−] reveals the missing part.

3. HANDS-ON LAB COACHING & STACK VERIFICATION KEYS (DOMAIN 02 WARM-UP B & INVESTIGATIONS 1–4)

Warm-Up B (Ext.): A number bond has a Whole of 8 and a known Part of 5. Find the missing Part 2. **Ans: Missing Part is 3 (8 - 5 = 3).**

Check Keys: [8] [ENTER] [5] [-]

Investigation 1: To make a Whole of 6, Part 1 is 2. What is the missing Part 2? **Ans: 4**

Check Keys: [6] [ENTER] [2] [-]

Investigation 2: In a number bond, Part 1 is 7 and Part 2 is 2. What is the Whole when you combine both parts? **Ans: 9**

Check Keys: [7] [ENTER] [2] [+]

Investigation 3: Three parts join together: first combine 2 and 3, then add 4 to that sum. What is the Whole? **Ans: 9**

Check Keys: [2] [ENTER] [3] [+] [4] [+]

Investigation 4: Ten birds sit on a fence. Seven fly to a tree. How many stay on the fence? **Ans: 3**

Check Keys: [1] [0] [ENTER] [7] [-]

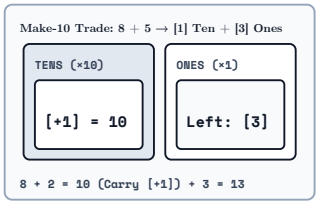
Complements to 10: Making Ten & Carry Trade

Core Concept Lesson • Big Ideas Math (Addition Strategies to 10) • Standard: CCSS.MATH.1.OA.C.6

1. MATHEMATICAL CONCEPT & STACK APPROACH: Determine complements to 10 using Place Value Digit Blocks, Gold [+1] Carry Tokens, and RPN subtraction.

Our base-10 system regroups every collection of 10 ones into 1 ten. Finding a number's complement to 10 (10 - 7 = 3) places the target base (10) on Shelf Y and the current ones digit (7) on Workbench X.

Classroom Hook: Place [8] in the Ones socket of your Place Value Tray. How many more ones do you need to trigger a Gold [+1] Carry Token in the Tens column?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Whole-Number Tray, Digit Blocks 0–9 & Gold [+1] Carry Tokens)

Hands-On Setup: Place the [7] Digit Block into the Ones socket of the Place Value Tray and hold a Gold [+1] Carry Token beside the Tens socket.

Predict First (Before Calculator): How many more ones must be added to [7] to reach 10 and trigger the Gold [+1] Carry Token? (7 + 3 = 10).

2A. TEACHER TALK — WHAT TO SAY & ASK:

"How many more to reach 10? Let's borrow 2 from 5 to complete 10 first and crown the Tens column with a Gold [+1] Carry Token!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Memorize 8 + 5 = 13 from your flash cards."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: `1` `0` `ENTER` `7` `-`

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
`1`	1	Enter 1
`0`	10	Target ten (10) in X
`ENTER`	10	10 lifted to Y
`7`	7	Current Ones digit (7) in X
`-`	3	10 - 7 = 3: Adding 3 completes 10 and triggers the [+1] Carry!

4. CONTEXT (Making a Ten First (8 + 5 = 10 + 3 = 13) & The 10-for-1 Carry Trade) • From Blocks to the Stack (SC-6 Manual) — Ch. 3: Making a Ten First & Trading 10 Ones + Appendix B: Station F.4

Why is 10 + 3 so much easier for our brains than 8 + 5? Because base-10 place value groups everything into full tens! When a column reaches 10 ones on the Place Value Tray, students trade all 10 ones for a single Gold [+1] Carry Token in the Tens column and leave the remaining ones in the Ones socket.

Core Investigation (Student Warm-Up A): The Ones socket holds [6]. How many more ones are needed to complete a full 10 and trade for 1 Ten?

Worked Answer: 4 ones (10 - 6 = 4). (All Keys: stackcalc.io/curriculum/answers#unit-03)

Verification Keys: `1` `0` `ENTER` `6` `-`

Bridging Through 10: The Make-Ten & Carry Strategy

Follow-Up Extension (Common Core / TEKS (Eureka, enVision, Bridges) • CCSS.MATH.1.OA.C.6) + Misconception Remediations & Lab Verification

1. FOLLOW-UP EXTENSION LESSON: CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Bridging across 10 decomposes the second addend so its first part completes a full ten (8 + 5 = (8 + 2) + 3 = 10 + 3). Completing the ten first turns any teen addition into an easy 10 + n fact.

Blocks & Prediction: Place [8] in Row 1 and [5] in Row 0 of the Ones column on the Place Value Tray. Prediction: Take 2 from the 5 to bring 8 up to 10. Place a Gold [+1] Carry Token in the Tens socket and [3] in the Ones socket: 10 + 3 = 13.

Extension Key Sequence: [8] [ENTER] [2] [+] [3] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[8]	8	8 in Ones socket
[ENTER]	8	8 lifted to Y
[2]	2	2 borrowed from 5 to make 10
[+]	10	8 + 2 = 10 (Gold [+1] Carry Token placed in Tens column!)
[3]	3	Remaining 3 ones
[+]	13	10 + 3 = 13 total ([1] Ten, [3] Ones)!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Counting on by ones across 10 (e.g., 8 + 5 as 9, 10, 11, 12, 13) instead of decomposing 5 into 2 + 3 to complete 10 first.

Hands-On Block & Stack Fix: Have the student bridge [8] to 10 on the Place Value Tray, place a Gold [+1] Carry Token in the Tens socket with [3] in the Ones socket, and key [8] [ENTER] [2] [+] [3] [+].

Misconception 2: Reading teen numbers as two separate single digits (seeing 14 as 1 and 4 rather than 1 Ten plus 4 Ones).

Hands-On Block & Stack Fix: Place [1] in the Tens socket and [4] in the Ones socket of the Place Value Tray and key [10] [ENTER] [4] [+] so students see 14 formed from 10 in Register Y and 4 in Register X.

3. HANDS-ON LAB COACHING & STACK VERIFICATION KEYS (DOMAIN 03 WARM-UP B & INVESTIGATIONS 1–4)

Warm-Up B (Ext.): Use the Make-Ten strategy on your Place Value Tray to add 9 + 6. What is the sum? **Ans: 15 (9 + 1 = 10, then 10 + 5 = 15).**

Check Keys: [9] [ENTER] [1] [+] [5] [+]

Investigation 1: You have 4 ones on your tray. How many more ones do you need to make a full 10? **Ans: 6**

Check Keys: [1] [0] [ENTER] [4] [-]

Investigation 2: Solve 7 + 5 by bridging through 10 on your Place Value Tray. **Ans: 12**

Check Keys: [7] [ENTER] [3] [+] [2] [+]

Investigation 3: The Ones socket holds [9]. How many more ones are needed to complete 10 and regroup into 1 Ten? **Ans: 1**

Check Keys: [1] [0] [ENTER] [9] [-]

Investigation 4: Solve 8 + 6 by bridging through 10 on your Place Value Tray. **Ans: 14**

Check Keys: [8] [ENTER] [2] [+] [4] [+]

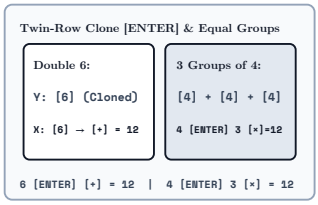
Twin-Row Doubles: Adding Two Equal Groups

Core Concept Lesson • Bridges in Mathematics (Doubles & Fact Strategies) • Standard: CCSS.MATH.1.OA.C.6

1. MATHEMATICAL CONCEPT & STACK APPROACH: Model doubles physically using matching digit blocks in Row 1 and Row 0 of the Place Value Tray and connect to equal groups.

Doubling a quantity means adding two identical copies of the same number ($n + n = 2n$). Seeing matching quantities in the top and bottom rows of the tray anchors mental doubles facts.

Classroom Hook: Place a [6] Digit Block in Row 1 and a matching [6] Digit Block in Row 0 of your Place Value Tray. Which single key clones a number on the calculator?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Two-Row Tray & Digit Blocks 0–9)

Hands-On Setup: Place a [6] Digit Block in Row 1 (top) and a matching [6] Digit Block in Row 0 (bottom) of the Ones column on the Place Value Tray.

Predict First (Before Calculator): Observe the matching rows: 6 + 6 = 12 (1 Ten and 2 Ones on the tray).

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Press [6] [ENTER]—look how [ENTER] clones your top row of 6 onto Shelf Y so both rows are ready to add!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Press 6, then plus, then 6, then equals."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [6] [ENTER] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[6]	6	Top row quantity (6) enters register X
[ENTER]	6	ENTER duplicates 6 into register Y! Both rows now on stack
[+]	12	Plus combines top and bottom rows: 6 + 6 = 12!

4. CONTEXT (Symmetry & Twin Rows: Why Doubles Anchor Mental Math) • From Blocks to the Stack (SC-6 Manual) — Ch. 1: Meet Your Math Workbench & Ch. 2: What Happens When You Press [ENTER]?

From two hands of 5 fingers (5 + 5 = 10) to two rows of 6 eggs in a carton (6 + 6 = 12), symmetry makes doubles the fastest facts in human memory. Once a student knows 7 + 7 = 14 by sight on the two-row tray, neighbor facts like 7 + 8 = 15 and 7 + 6 = 13 become instant one-step adjustments.

Core Investigation (Student Warm-Up A): Place [7] in both rows of your Place Value Tray to build a twin-row double. What is the sum of two 7*s*?

Worked Answer: 14 ([1] Ten and [4] Ones on the tray; 7 + 7 = 14). (All Keys: stackcalc.io/curriculum/answers#unit-04)

Verification Keys: [7] [ENTER] [+]

Near-Doubles: Doubles Plus One

Follow-Up Extension (Bridges in Mathematics • CCSS.MATH.2.OA.B.2) + Misconception Remediations & Lab Verification

1. FOLLOW-UP EXTENSION LESSON: CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: A near-double (such as 7 + 8) is a double plus or minus 1 ((7 + 7) + 1). Evaluating the known double first (14) makes adding the remaining +1 effortless.

Blocks & Prediction: Place [7] in Row 1 (top) and [8] in Row 0 (bottom) of the Place Value Tray. Prediction: 7 and 8 is double 7 plus 1 more: 14 + 1 = 15.

Extension Key Sequence: [7] [ENTER] [+] [1] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[7]	7	Enter base number 7
[ENTER]	7	Duplicate 7
[+]	14	Double: 7 + 7 = 14
[1]	1	Near-double difference (1)
[+]	15	14 + 1 = 15!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Retyping the number to double it (e.g., [7] [ENTER] [7] [+]) instead of recognizing that [ENTER] duplicates Register X into Register Y.

Hands-On Block & Stack Fix: Place matching [7] Digit Blocks in Row 1 and Row 0 of the Place Value Tray, then press [7] [ENTER] and point to the screen: both Y and X already hold 7, so pressing [+] doubles it immediately.

Misconception 2: Confusing '3 groups of 4' with '3 + 4' when transitioning from repeated addition to multiplication.

Hands-On Block & Stack Fix: Model three groups of [4] on the Place Value Tray; have the student key [4] [ENTER] [4] [+] [4] [+] = 12 first, then compare directly with [4] [ENTER] [3] [×] = 12.

3. HANDS-ON LAB COACHING & STACK VERIFICATION KEYS (DOMAIN 04 WARM-UP B & INVESTIGATIONS 1–4)

Warm-Up B (Ext.): Use the known double 6 + 6 to solve the near-double 6 + 7. **Ans: 13 (6 + 6 = 12, then 12 + 1 = 13).**

Check Keys: [6] [ENTER] [+] [1] [+]

Investigation 1: Two matching rows on the Place Value Tray each hold 8. What is double 8? **Ans: 16**

Check Keys: [8] [ENTER] [+]

Investigation 2: Use the double 8 + 8 to solve the near-double 8 + 9. **Ans: 17**

Check Keys: [8] [ENTER] [+] [1] [+]

Investigation 3: A carton has two equal rows of 9 eggs. How many eggs are in the carton? **Ans: 18**

Check Keys: [9] [ENTER] [+]

Investigation 4: Use the double 5 + 5 to solve the near-double 5 + 6. **Ans: 11**

Check Keys: [5] [ENTER] [+] [1] [+]

Multi-Step Word Problems Without Parentheses

Core Concept Lesson • Common Core / TEKS (Eureka, Illustrative, enVision) (Multi-Step Word Problems) • Standard: CCSS.MATH.4.OA.A.3

1. MATHEMATICAL CONCEPT & STACK APPROACH: Evaluate chained arithmetic expressions without parentheses by managing stack register flow.

In a multi-step expression like $(3 \times 4) + (2 \times 5)$, each product is a self-contained quantity that must be evaluated before the two groups can be added. On the 4-level stack, the first product (12) waits safely on Shelf Y while the second product (10) is formed on X, requiring zero parentheses.

Classroom Hook: Beat the Smartphone Challenge: Ask a volunteer to type 8 + 4 × 5 left-to-right on a basic phone calculator (it says 60!). Why is the true answer 28?



Place Value Studio • 10-for-1 Carry Trade

2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Place Value Blocks & 4-Register Stack

Hands-On Setup: Use Place Value unit blocks to represent quantities alongside the 4-line stack display.

Predict First (Before Calculator): Problem: (8 - 3) × 2. Predict: 8 - 3 = 5, then 5 × 2 = 10.

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Solve the inner bundle first and let it wait on Shelf Y; when you type the next number, watch Automatic Stack Lift push it up!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Type the expression from left to right and hope the calculator knows PEMDAS."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [8] [ENTER] [3] [-] [2] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[8]	8	8 in X
[ENTER]	8	8 copied to Y
[3]	3	3 entered into X
[-]	5	8 - 3 = 5 in X
[2]	2	Automatic lift! 5 lifts to Y; 2 enters X
[×]	10	5 × 2 = 10 in X!

4. CONTEXT (Inside-Out Multi-Step Problem Solving & Young Gauss's Pairing Insight) • From Blocks to the Stack (SC-6 Manual) — Ch. 2: The Number Elevator & Ch. 4: Young Gauss's 10-Second Shortcut

In 1786, 9-year-old Carl Friedrich Gauss stunned his teacher by adding 1 through 100 in seconds—grouping the numbers from the outside in into 50 pairs of 101! Whenever we solve a multi-step story problem on paper, we naturally compute the inner group first and hold that intermediate result aside while we set up the next step.

Core Investigation (Student Warm-Up A): A baker bakes 15 cookies, sells 6 of them, and packs the remaining cookies equally into 3 bags. How many cookies go in each bag?

Worked Answer: 3 cookies per bag ((15 - 6) ÷ 3 = 3). (All Keys: stackcalc.io/curriculum/answers#unit-05)

Verification Keys: [1] [5] [ENTER] [6] [-] [3] [÷]

Automatic Stack Lift & Register Retention

Follow-Up Extension (Big Ideas Math • CCSS.MATH.4.OA.A.3) + Misconception Remediations & Lab Verification

1. FOLLOW-UP EXTENSION LESSON: CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: When an inner operation finishes, its result becomes the starting quantity for the next step. On the stack, that intermediate result automatically lifts to Waiting Shelf Y as soon as the next number is typed—allowing multi-step math like (12 ÷ 4) + 7 to flow continuously from the inside out.

Blocks & Prediction: Observe the 4-line stack screen as values automatically lift into register Y: (12 / 4) + 7. Prediction: 12 / 4 = 3. Entering 7 will lift 3 to Y. 3 + 7 = 10.

Extension Key Sequence: [1] [2] [ENTER] [4] [÷] [7] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	First digit of 12 enters X
[2]	12	12 in register X
[ENTER]	12	12 copied to Y
[4]	4	4 enters X
[÷]	3	12 / 4 = 3 in X
[7]	7	7 entered: 3 lifts to Y automatically!
[+]	10	3 + 7 = 10!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Pressing [ENTER] after an operator (+, −, ×, ÷) before typing the next number in a chained calculation, wasting a stack register.

Hands-On Block & Stack Fix: Demonstrate Automatic Stack Lift: after [18] [ENTER] [6] [−] produces 12, typing [3] automatically lifts 12 to Register Y—no [ENTER] needed before [3] [×]!

Misconception 2: Reversing operand order in subtraction or division (computing $X - Y$ instead of $Y - X$).

Hands-On Block & Stack Fix: Remind students that Register Y is always the starting quantity on top (Row 1) and Register X is the operator's action (Row 0); if entered backward, press [x swap y] before [−] or [÷].

3. HANDS-ON LAB COACHING & STACK VERIFICATION KEYS (DOMAIN 05 WARM-UP B & INVESTIGATIONS 1–4)

Warm-Up B (Ext.): Evaluate (24 ÷ 6) + 15 by solving the division group first and then adding 15 to that result. **Ans: 19 (24 ÷ 6 = 4, then 4 + 15 = 19).**

Check Keys: [2] [4] [ENTER] [6] [÷] [1] [5] [+]

Investigation 1: Solve (14 - 5) × 3 by finding the difference inside the parentheses first, then multiplying by 3. **Ans: 27**

Check Keys: [1] [4] [ENTER] [5] [-] [3] [×]

Investigation 2: Solve 30 ÷ (2 + 3) by adding the divisor group (2 + 3) first, then dividing 30 by that sum. **Ans: 6**

Check Keys: [2] [ENTER] [3] [+] [3] [0] [↑↓] [÷]

Investigation 3: Evaluate (9 + 7) ÷ (8 - 4) by finding the numerator sum and denominator difference first, then dividing. **Ans: 4**

Check Keys: [9] [ENTER] [7] [+] [8] [ENTER] [4] [-] [÷]

Investigation 4: A farmer collects 20 eggs, sets aside 2 cracked eggs, and divides the rest equally among 6 cartons. How many per carton? **Ans: 3 eggs**

Check Keys: [2] [0] [ENTER] [2] [-] [6] [÷]

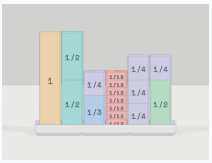
Mixed Numbers & Height Equivalence: 3/4 + 2 1/2

Core Concept Lesson • Common Core / TEKS (Eureka, Illustrative, Big Ideas) (Fraction Addition & Mixed Numbers) • Standard: CCSS.MATH.5.NF.A.1

1. MATHEMATICAL CONCEPT & STACK APPROACH: Represent proper and mixed fractions physically with block towers and computationally with two-dot notation.

Fractions can only be added or subtracted when they share a common unit (denominator), and mixed numbers combine whole units with fractional parts. On the stack, two-dot notation ([.] n [.] d) treats each fraction or mixed number as a single unified value on Y and X.

Classroom Hook: Hold up the 1-Whole tower and a millimeter ruler: Why did engineers make the Whole block exactly 84 mm tall instead of 100 mm?



Fraction Studio • 84 mm Flush Towers

2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: 84 mm Fraction Towers & Twin Comparator Well

Hands-On Setup: Place three 1/4 blocks (3 × 21 = 63 mm). Beside them, stack two Whole blocks and one 1/2 block (84 + 84 + 42 = 210 mm).

Predict First (Before Calculator): Combine heights: 63 mm + 210 mm = 273 mm. In wholes: 3 1/4 (273 / 84 = 3.25).

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Stand your 1/2 (42 mm) and 1/4 (21 mm) towers in the comparator lid—do they match three 1/4 towers (63 mm) in height?"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Just cross-multiply the denominators and don't ask why."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [•] [3] [•] [4] [ENTER] [2] [•] [1] [•] [2] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[•]	0.	Leading dot indicates 0 wholes
[3]	0.3	Numerator = 3
[•]	0.3.	Second dot sets fraction entry mode
[4]	3/4	Denominator = 4: displays 3/4 (or 0.75)
[ENTER]	3/4	3/4 lifted to register Y
[2]	2	Whole = 2
[•]	2.	Dot
[1]	2.1	Numerator = 1
[•]	2.1.	Dot
[2]	2 1/2	Denominator = 2: displays 2 1/2
[+]	3 1/4	Plus combines: 3 1/4 (decimal 3.25)!

4. CONTEXT (The Pharaoh's Bakery & Why the Whole Fraction Tower Stands 84.0 mm Tall) • From Blocks to the Stack (SC-6 Manual) — Ch. 5: Fractions, Decimals & Ratios (The Pharaoh's Bakery & 84mm Comparator)

Four thousand years ago, Egyptian scribes divided 3 loaves of bread fairly among 4 workers by giving each worker 1/2 + 1/4 of a loaf! Because the Whole block in the Fraction Studio is engineered to 84.0 mm (divisible by 1, 2, 3, 4, 6, 7, 8, and 12), a 42 mm (1/2) block plus a 21 mm (1/4) block stacks to 63 mm—matching three 21 mm (3/4) blocks by sight and touch.

Core Investigation (Student Warm-Up A): Stack a 1/2 tower and a 1/4 tower together. How many 1/4 blocks match their combined height, and what is 1/2 + 1/4?

Worked Answer: 3/4 (three 21 mm fourth-blocks = 63 mm). (All Keys: stackcalc.io/curriculum/answers#unit-06)

Verification Keys: [•] [1] [•] [2] [ENTER] [•] [1] [•] [4] [+]

Fraction Multiplication & Reciprocals

Follow-Up Extension (Bridges in Mathematics • CCSS.MATH.5.NF.B.4) + Misconception Remediations & Lab Verification

1. FOLLOW-UP EXTENSION LESSON: CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Multiplying (3/4) × (2/3) scales one tower by another—showing unreduced 6/12 first before [SIMP] reduces to 1/2! For fraction division (1/2 ÷ 3/4), use Split & Rebuild (with a 4-kit table pod): split 1/2 (42 mm) into 3 equal 1/6 blocks (14 mm each, ÷3), then stack 4 of them (56 mm = 2/3, ×4) to rebuild 1 full tower!

Blocks & Prediction: Set up 3/4 of a tower (63 mm) and divide into 3 equal parts (21 mm each), taking 2 parts: (3/4) × (2/3). Prediction: 3/4 × 2/3 = 6/12 = 1/2. The resulting block height matches one 1/2 block (42 mm).

Extension Key Sequence: [•] [3] [•] [4] [ENTER] [•] [2] [•] [3] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[•]	0.	Start proper fraction 3/4
[3]	0.3	Numerator 3
[•]	0.3.	Fraction separator
[4]	3/4	3/4 in register X
[ENTER]	3/4	3/4 lifted to register Y
[•]	0.	Start proper fraction 2/3
[2]	0.2	Numerator 2
[•]	0.2.	Fraction separator
[3]	2/3	2/3 in register X
[×]	6/12→1/2	Shows 6/12 first; tap [SIMP] to reduce to 1/2 (42 mm)!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Adding numerators and denominators directly (e.g., 1/2 + 1/4 = 2/6) or treating 'invert and multiply' for fraction division as an arbitrary rule.

Hands-On Block & Stack Fix: In the comparator lid, 1/2 (42 mm) + 1/4 (21 mm) matches three 1/4 blocks (63 mm), not two 1/6 blocks (28 mm). For division (e.g., 1/2 ÷ 3/4 = 2/3), have a 4-kit table pod combine blocks to Split & Rebuild 1 Full Tower: split 1/2 (42 mm) into three 1/6 blocks (÷3 → 14 mm is 1/4 of the mystery tower), then stack four 1/6 blocks (×4 → 56 mm = two 1/3 blocks) to prove ÷(3/4) = ×(4/3).

Misconception 2: Entering a proper fraction like 3/4 as [3] [.] [4] (which is decimal 3.4) instead of using Two-Dot fraction notation [.][3][.][4].

Hands-On Block & Stack Fix: Teach the rule: 'First dot = whole number zero, second dot = fraction bar.' Typing [.][3][.][4] displays exact 3/4 on the stack.

3. HANDS-ON LAB COACHING & STACK VERIFICATION KEYS (DOMAIN 06 WARM-UP B & INVESTIGATIONS 1–4)

Warm-Up B (Ext.): Find 2/3 of 3/8 by multiplying (2/3) × (3/8) and simplifying to lowest terms. Which unit fraction tower matches the product? **Ans: 1/4 (6/24 = 1/4, height 21 mm).**

Check Keys: [•] [2] [•] [3] [ENTER] [•] [3] [•] [8] [×]

Investigation 1: Add 1/3 + 1/6 and simplify the sum to lowest terms. **Ans: 1/2 (42 mm)**

Check Keys: [•] [1] [•] [3] [ENTER] [•] [1] [•] [6] [+]

Investigation 2: Subtract 1 1/2 - 3/4. What fraction remains? **Ans: 3/4 (63 mm)**

Check Keys: [1] [•] [1] [•] [2] [ENTER] [•] [3] [•] [4] [-]

Investigation 3: Compare 3/8 to 1/2. Which is smaller, and what is 3/8 written as a decimal? **Ans: 0.375**

Check Keys: [•] [3] [•] [8] [FDISP]

Investigation 4: Find the area of a rectangle measuring 2/3 unit by 3/4 unit by multiplying (2/3) × (3/4) and simplifying with [SIMP]. **Ans: 6/12 → 1/2 (42 mm)**

Check Keys: [•] [2] [•] [3] [ENTER] [•] [3] [•] [4] [×] [g] [SIMP]

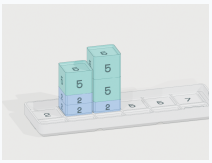
Chained Operations: Evaluating (a + b) × (c - d) from the Inside Out

Core Concept Lesson • Big Ideas Math (Numerical & Algebraic Expressions) • Standard: CCSS.MATH.6.EE.A.2

1. MATHEMATICAL CONCEPT & STACK APPROACH: Evaluate multi-branch expressions from the inside out using prime factor towers and verify on the 4-register stack without parentheses.

Written algebra uses parentheses to show which inner operations must happen first. Evaluating from the inside out follows true mathematical structure: compute the first branch (6 + 1 = 7), hold that result while computing the second branch (9 - 5 = 4), and multiply the two branch values together.

Classroom Hook: Why does 24 factor into the Prime Factor tower [2]×[2]×[2]×[3] (giving 8 factor pairs), while 23 is a single prime number?



Prime Factor Studio • Log Prime Towers

2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Prime Factor Studio (Logarithmic Prime Blocks) & 4-Register Stack

Hands-On Setup: Use Prime Factor Studio blocks in the 12-socket comparator lid to model the two factor branches: (6 + 1) = [7] prime block and (9 - 5) = [2] + [2] = 4 tower, then combine their heights for 7 × 4 = 28.

Predict First (Before Calculator): Left sub-expression: 6 + 1 = 7. Right sub-expression: 9 - 5 = 4. Combined product: 7 × 4 = 28 (prime tower 2 × 2 × 7).

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Evaluate the left branch (6 + 1 = 7) onto Shelf Y and the right branch (9 − 5 = 4) on Shelf X, then press [×] to combine!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Count open and close parentheses before typing anything."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: 6 ENTER 1 + 9 ENTER 5 - ×

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
6	6	Value a (6) placed into Register X
ENTER	6	ENTER lifts 6 into Register Y
1	1	Value b (1) placed into Register X
+	7	Left operator (+): Combines Y(6) + X(1) = 7 in X
9	9	Value c (9) entered into X; 7 lifted to Y
ENTER	9	ENTER lifts 9 into Z and 7 into T
5	5	Value d (5) placed into Register X
-	4	Right operator (-): Combines Z/ Y(9) - X(5) = 4 in X; 7 drops to Y
×	28	Root operator (×): Combines Y(7) × X(4) = 28 in X!

4. CONTEXT (Why 24 Is the Champion Packing Number & How Inside-Out Evaluation Eliminates Parentheses) • From Blocks to the Stack (SC-6 Manual) — Ch. 7: Prime Numbers & Factor Towers + Thinking in the Stack (SC-32 Manual) — Prologue, Ch. 2: Inside-Out Thinking & Ch. 12: Coordinate Bridge

Why are eggs sold by the dozen and hours divided into 24? Because 24 = 2 × 2 × 2 × 3 has eight divisors (1, 2, 3, 4, 6, 8, 12, 24), making it a champion packing number! And when evaluating nested expressions, 16th-century typesetters only invented parentheses to flatten 2D fraction bars onto a single line of type—whereas evaluating from the inside out resolves every branch naturally.

Core Investigation (Student Warm-Up A): Evaluate (8 + 2) × (14 - 9) by finding the value of each parenthesized group first, then multiplying the two results.

Worked Answer: (8 + 2) = 10 and (14 - 9) = 5; 10 × 5 = 50 (Prime factors: 2 × 5 × 5). (All Keys: stackcalc.io/curriculum/answers#unit-07)

Verification Keys: 8 ENTER 2 + 1 4 ENTER 9 - ×

Nested Fractions & Multi-Layer Expressions

Follow-Up Extension (Common Core / TEKS (Illustrative, Eureka, Big Ideas) • CCSS.MATH.6.EE.A.2) + Misconception Remediations & Lab Verification

1. FOLLOW-UP EXTENSION LESSON: CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Deeply nested fractions such as (12 - 4) ÷ (2 × (1 + 1)) require evaluating the numerator (8) and the innermost group of the denominator (1 + 1 = 2, then 2 × 2 = 4) before dividing. The 4-register stack (T, Z, Y, X) mirrors this exact paper-and-pencil hierarchy.

Blocks & Prediction: Model the numerator (12 - 4 = 8, three [2] blocks) and denominator (2 × (1 + 1) = 4, two [2] blocks) with Prime Factor towers. Prediction: Numerator: 12 - 4 = 8 (three [2] blocks). Denominator: 2 × 2 = 4 (two [2] blocks). Result: 8 ÷ 4 = 2.

Extension Key Sequence: [1] [2] [ENTER] [4] [-] [2] [ENTER] [1] [ENTER] [1] [+] [×] [÷]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	First digit of 12
[2]	12	12 in register X
[ENTER]	12	12 lifted to Y
[4]	4	4 in register X
[-]	8	Numerator 12 - 4 = 8 held in X
[2]	2	Factor 2 enters X; 8 lifts to Y
[ENTER]	2	2 lifted to Y; 8 lifts to Z
[1]	1	Inner 1 enters X
[ENTER]	1	1 lifted to Y; 2 in Z; 8 in T
[1]	1	Second 1 enters X
[+]	2	Innermost sum: 1 + 1 = 2 in X
[×]	4	Denominator combined: 2 × 2 = 4 in X; 8 in Y
[÷]	2	8 / 4 = 2!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Evaluating expressions strictly left-to-right (computing 10 + 5 × 2 as 15 × 2 = 30) instead of respecting operator structure.

Hands-On Block & Stack Fix: Model the product branch 5 × 2 with Prime Factor [5] and [2] blocks first: push 10 to Register Y ([10] [ENTER]), evaluate 5 × 2 in X ([5] [ENTER] [2] [×] → 10), then press [+] → 20.

Misconception 2: Losing track of intermediate results when evaluating two parenthesized branches like (8 + 2) × (9 − 4).

Hands-On Block & Stack Fix: Show how Register Y holds the first branch (8 + 2 = 10) automatically while the second branch (9 − 4 = 5) is computed in X, ready for [×] = 50.

3. HANDS-ON LAB COACHING & STACK VERIFICATION KEYS (DOMAIN 07 WARM-UP B & INVESTIGATIONS 1–4)

Warm-Up B (Ext.): Evaluate (15 - 5) ÷ (3 + 2) by computing the numerator and denominator separately, then dividing.

Ans: Numerator = 10, Denominator = 5; 10 ÷ 5 = 2.

Check Keys: [1] [5] [ENTER] [5] [-] [3] [ENTER] [2] [+] [÷]

Investigation 1: Evaluate (4 + 3) × (10 - 6) by solving each parenthesized group first.

Ans: 7 × 4 = 28

Check Keys: [4] [ENTER] [3] [+] [1] [0] [ENTER] [6] [-] [×]

Investigation 2: Evaluate (20 - 4) ÷ (2 × 4) by finding the numerator and denominator first.

Ans: 16 ÷ 8 = 2

Check Keys: [2] [0] [ENTER] [4] [-] [2] [ENTER] [4] [×] [÷]

Investigation 3: Evaluate (18 - 6) ÷ (1 + 2) and write the prime factorization of the numerator.

Ans: 12 ÷ 3 = 4 (12 = 2×2×3)

Check Keys: [1] [8] [ENTER] [6] [-] [1] [ENTER] [2] [+] [÷]

Investigation 4: Evaluate 5 + (3 × 4) by computing the inner product before adding 5.

Ans: 5 + 12 = 17

Check Keys: [3] [ENTER] [4] [×] [5] [+]

Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ with Prime Factor Blocks

Core Concept Lesson • Big Ideas Math (Polynomial Equations & Factoring) • Standard: CCSS.MATH.HSA.APR.A.1

1. MATHEMATICAL CONCEPT & STACK APPROACH: Factor the constant term with Prime Factor blocks, sketch the (x + p)(x + q) area grid, and find the roots of the quadratic equation.

A polynomial $ax^2 + bx + c$ mirrors base-10 place value with variable powers (x^2, x^1, x^0) in place of 10^2, 10^1, 10^0. Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ decomposes the constant term into prime factors ($2 \times 3 = 6$) whose sum equals the linear coefficient ($2 + 3 = 5$).

Classroom Hook: How is the polynomial $x^2 + 5x + 6 = (x + 2)(x + 3)$ just like the base-10 number $156 = 12 \times 13$ when $x = 10$?

Area Grid $(x+2)(x+3)$ & Prime Pair 2×3=6

x^2	$+3x$
$+2x$	$2 \times 3 = 6$

$b = 2 + 3 = 5$, $c = 2 \times 3 = 6 \rightarrow$ Roots -2, -3

2. MODELING WITH BLOCKS & STUDENT PREDICTION

Block Kit & Mathematical Focus: Prime Factor Studio & Algebraic Area Diagram

Hands-On Setup: Stack Prime Factor blocks [2] and [3] ($2 \times 3 = 6$) in the comparator lid and sketch the 2×2 area grid for $(x + 2)(x + 3) = x^2 + 3x + 2x + 6$.

Predict First (Before Calculator): Prime factors of 6 are 2 and 3, and their sum is $2 + 3 = 5$. The factored form is $(x + 2)(x + 3)$ with roots $x = -2$ and $x = -3$.

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Factor the constant 6 with Prime Factor blocks [2] and [3]—their sum $2 + 3 = 5$ matches the middle x^1 coefficient, giving factors $(x + 2)(x + 3)$!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Just FOIL the letters and memorize the quadratic formula."

3. CALCULATOR EXECUTION: SC-32 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [g:] [POLY] 1 [B] 5 [C ON] 6 [D] [XEQ]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[g:] [POLY]	0	Press [g:] [POLY] to open $Ax^3 + Bx^2 + Cx + D = 0$ (with $A = 0$)
1 [B]	1	Key 1 and tap softkey [B] for $B = 1$ (x^2 coefficient)
5 [C ON]	5	Key 5 and tap softkey [C] for $C = 5$ (x coefficient)
6 [D]	6	Key 6 and tap softkey [D] for $D = 6$ (constant term)
[XEQ]	-2	Press [XEQ]: displays r1: -3 (factor x + 3) and r2: -2 (factor x + 2)!

4. CONTEXT (Al-Khwarizmi's Geometric Squares & The 1535 Public Algebra Duel in Milan) • Thinking in the Stack (SC-32 Manual) — Ch. 6: Galileo's Square-Cube Area Law + Ch. 7: Polynomial Evaluation, Cardano's Cubic & Brent's Root-Finder ([POLY] & [SOLVE]) + Ch. 8: Romberg Calculus

In 820 AD in Baghdad, Al-Khwarizmi solved $x^2 + 10x = 39$ by drawing literal geometric squares and rectangles ('completing the square')—and in 1535 in Milan, Niccolò Tartaglia defeated Antonio Fiore in a 30-problem public algebra duel overnight! By sketching a 2×2 area grid and factoring constant terms with Prime Factor blocks, students see why (x + p)(x + q) works geometrically.

Core Investigation (Student Warm-Up A): Factor $x^2 + 7x + 12$ by decomposing 12 with Prime Factor blocks to find the factor pair whose sum is 7, and find both roots of $x^2 + 7x + 12 = 0$.

Worked Answer: $3 \times 4 = 12$ and $3 + 4 = 7$; factored form $(x + 3)(x + 4)$, roots $x = -3$ and $x = -4$. (All Keys: stackcalc.io/curriculum/answers#unit-08)

Verification Keys: [g:] [POLY] 1 [B] 7 [C ON] 1 2 [D] [XEQ]

Signed Factor Pairs: Factoring Trinomials with Negative Terms

Follow-Up Extension (Singapore Math • CCSS.MATH.HSA.REI.B.4) + Misconception Remediations & Lab Verification

1. FOLLOW-UP EXTENSION LESSON: CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: When a quadratic has a negative constant or linear term (such as $x^2 - x - 6$), the factor pair must have opposite signs (-3 and +2) so their product is -6 and their sum is -1. You can check both roots on the calculator after factoring on paper.

Blocks & Prediction: Factor |c| = 6 into Prime Factor blocks [2] and [3] and sketch a 2×2 algebraic area grid for $x^2 - x - 6$. Prediction: Assign signs to the factor pair (2, -3) so their product is -6 and their sum is 2 + (-3) = -1: factors are $(x - 3)(x + 2)$.

Extension Key Sequence: [g] [POLY] 1 [B] 1 [+/−] [C ON] 6 [+/−] [D] [XEQ]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[g] [POLY]	0	Press [g:] [POLY] to open $Ax^3 + Bx^2 + Cx + D = 0$ (with $A = 0$)
1 [B]	1	Store $B = 1$ (x^2 coefficient)
1 [+/−] [C ON]	-1	Store C = -1 (x coefficient)
6 [+/−] [D]	-6	Store D = -6 (constant term)
[XEQ]	3	Press [XEQ]: displays r1: -2 (factor x + 2) and r2: 3 (factor x - 3)!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Expanding $(x + 2)(x + 3)$ as $x^2 + 6$ by forgetting the two linear cross-terms ($2x$ and $3x$).

Hands-On Block & Stack Fix: Sketch a 2×2 algebraic area grid for $(x + 2)(x + 3)$ and factor 6 into Prime Factor blocks [2] and [3] so students see that $b = 2 + 3 = 5$ and $c = 2 \times 3 = 6$.

Misconception 2: Using the subtraction operator [−] instead of the change-sign key [+/−] when entering negative polynomial coefficients into [POLY].

Hands-On Block & Stack Fix: Point out that [−] immediately subtracts X from Y, whereas [+/−] negates the number currently in Register X without collapsing the stack.

3. HANDS-ON LAB COACHING & STACK VERIFICATION KEYS (DOMAIN 08 WARM-UP B & INVESTIGATIONS 1–4)

Warm-Up B (Ext.): Factor $x^2 - 5x + 6$ into two binomials by finding two negative factors of +6 whose sum is -5, and state both roots.
Ans: (-2) × (-3) = +6 and (-2) + (-3) = -5; factored form $(x - 2)(x - 3)$, roots $x = 2$ and $x = 3$.
Check Keys: [g] [POLY] 1 [B] 5 [+/−] [C ON] 6 [D] [XEQ]

Investigation 1: Factor $x^2 + 6x + 8$ by decomposing 8 into a factor pair that sums to 6, and find the roots.
Ans: (x + 2)(x + 4); roots -2, -4
Check Keys: [g] [POLY] 1 [B] 6 [C ON] 8 [D] [XEQ]

Investigation 2: Factor $x^2 - 4$ (difference of two squares, where the linear coefficient $b = 0$).
Ans: (x - 2)(x + 2); roots 2, -2
Check Keys: [g] [POLY] 1 [B] 0 [C ON] 4 [+/−] [D] [XEQ]

Investigation 3: Factor $x^2 + 5x + 4$ into two binomials and state the solutions to $x^2 + 5x + 4 = 0$.
Ans: (x + 1)(x + 4); roots -1, -4
Check Keys: [g] [POLY] 1 [B] 5 [C ON] 4 [D] [XEQ]

Investigation 4: Factor $x^2 - 5x + 6$ using signed factor pairs and state both roots.
Ans: (x - 2)(x - 3); roots 2, 3
Check Keys: [g] [POLY] 1 [B] 5 [+/−] [C ON] 6 [D] [XEQ]

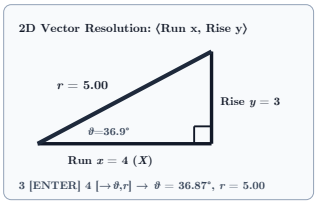
Vector Magnitude & Direction: Rectangular to Polar Coordinates

Core Concept Lesson • Big Ideas Math (Right Triangle Trigonometry & Vectors) • Standard: CCSS.MATH.HSN.VM.A.1

1. MATHEMATICAL CONCEPT & STACK APPROACH: Decompose 2D vectors into horizontal and vertical components (dx, dy) and calculate polar magnitude r and direction angle ϑ.

Every 2D vector or right triangle relates two perpendicular legs (dx, dy) to a hypotenuse r and angle ϑ via $r = \sqrt{x^2 + y^2}$ and $\tan(\vartheta) = y/x$. Placing vertical component dy in Register Y and horizontal component dx in Register X matches the key legend $[\rightarrow\vartheta,r]$ (placing ϑ in Y and r in X).

Classroom Hook: If a surveyor walks 4 blocks East and 3 blocks North, why is the straight-line shortcut 5 blocks instead of 4 + 3 = 7?



2. MATHEMATICAL CONCEPT & STUDENT PREDICTION

Mathematical Focus: Right-Triangle & 2D Vector Diagram

Mathematical Setup: Draw a right triangle with vertical rise $dy = 3$ (Register Y) and horizontal run $dx = 4$ (Register X) to model the 3-4-5 vector.

Predict First (Before Calculator): Rise = 3, Run = 4. Predict magnitude: $r = \sqrt{4^2 + 3^2} = 5$. Predict angle: arctan(3/4) ≈ 36.87°.

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Put your vertical Rise (Δy) on Waiting Shelf Y and horizontal Run (Δx) on Workbench X, then press $[\rightarrow\vartheta,r]$ to see hypotenuse r in X and angle ϑ in Y!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Type arctan(y/x) and guess which quadrant you're in."

3. CALCULATOR EXECUTION: SC-32 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: 3 ENTER 4 ↶ ▸θ,r x⇄y

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
3	3	Enter vertical component $y = 3$
ENTER	3	$y = 3$ lifted to Waiting Shelf Y
4	4	Enter horizontal component $x = 4$ in Workbench X
↶ ▸θ,r	5.00	Rectangular to Polar! Screen X displays magnitude $r = 5.00$ ($\vartheta = 36.87°$ in Y)
x⇄y	36.87	Swap registers: X displays direction angle $\vartheta = 36.87°$!

4. CONTEXT (A Stick in Alexandria: How Eratosthenes Measured the Earth with a Shadow) • Thinking in the Stack (SC-32 Manual) — Ch. 4: Circular Trigonometry & Vector Projections ($[\rightarrow\vartheta,r]$, $[\rightarrow y,x]$) + Ch. 5: Complex Numbers & AC Phasors + Ch. 6: Solid Geometry

In 240 BC, Eratosthenes measured a 7.2° shadow angle in Alexandria at noon on the summer solstice while a deep well in Syene 800 km away had zero shadow—calculating Earth's circumference within 1% using pure right-triangle ratios! That same relationship between perpendicular components (dx, dy) and polar angle/distance (r, ϑ) guides modern navigation and engineering.

Core Investigation (Student Warm-Up A): A surveyor walks 60 m East ($x = 60$) and 80 m North ($y = 80$). Use the Pythagorean theorem and tangent ratio to find the straight-line distance r and angle ϑ north of east.

Worked Answer: $r = \sqrt{60^2 + 80^2} = 100$ m; $\vartheta = \arctan(80/60) \approx 53.13°$. (All Keys: stackcalc.io/curriculum/answers#unit-09)

Verification Keys: 8 0 ENTER 6 0 ↶ ▸θ,r

2D Vector Addition: Resultant Forces by Component Summation

Follow-Up Extension (Big Ideas Math • CCSS.MATH.HSN.VM.A.3) + Misconception Remediations & Lab Verification

1. FOLLOW-UP EXTENSION LESSON: CONCEPT → PAPER STEPS → CALCULATOR

Concept & Stack Approach: Non-collinear vectors cannot be added by simply summing their lengths; their vertical (y) and horizontal (x) components must be summed separately first. Summing the *y*-components into Y and *x*-components into X prepares $\langle x_{net}, y_{net} \rangle$ directly for [→ϑ,r].

Mathematical Setup & Prediction: Sketch component vectors F1 = ⟨3, 2⟩ and F2 = ⟨1, 5⟩ tip-to-tail on a coordinate grid. Prediction: Add components: $y_{net} = 2 + 5 = 7$; $x_{net} = 3 + 1 = 4$. Magnitude $r = \sqrt{4^2 + 7^2} = \sqrt{65} \approx 8.06$.

Extension Key Sequence: [2] [ENTER] [5] [+] [3] [ENTER] [1] [+] [↶] [▸θ,r]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[2]	2	Vertical component y1 = 2
[ENTER]	2	Lifted to Y
[5]	5	Vertical component y2 = 5
[+]	7	Net vertical force $y_{net} = 7$ in X
[3]	3	Horizontal component x1 = 3 (7 lifts to Y)
[ENTER]	3	3 lifted to Y; 7 in Z
[1]	1	Horizontal component x2 = 1 in X
[+]	4	Net horizontal force $x_{net} = 4$ in X; $y_{net} = 7$ waits in Y
[↶] [▸θ,r]	8.06	Resultant force magnitude $r = 8.06$ in X (angle $\vartheta = 60.26°$ in Y)!

2. TARGET MISCONCEPTIONS & STACK REMEDIATIONS

Misconception 1: Evaluating trigonometric functions in Radian mode when angles are given in Degrees (or vice versa).

Mathematical & Stack Fix: Have students check the DEG/RAD indicator on the SC-32 status bar and verify on a 30°-60°-90° right-triangle sketch that sin(30°) = 0.5000 before starting vector problems.

Misconception 2: Swapping x and y components when converting rectangular coordinates to polar via [→ϑ,r].

Mathematical & Stack Fix: Label vertical Rise (dy) for Waiting Shelf Y and horizontal Run (dx) for Workbench X on your vector sketch before pressing [→ϑ,r].

3. HANDS-ON LAB COACHING & STACK VERIFICATION KEYS (DOMAIN 09 WARM-UP B & INVESTIGATIONS 1–4)

Warm-Up B (Ext.): Two cables pull on a post: Vector A is ⟨5, 2⟩ and Vector B is ⟨-2, 4⟩. Find the resultant vector $\langle x_{net}, y_{net} \rangle$ and its magnitude r.

Ans: Resultant ⟨3, 6⟩; Magnitude $r = \sqrt{3^2 + 6^2} = \sqrt{45} \approx 6.71$.

Check Keys: [2] [ENTER] [4] [+] [5] [ENTER] [2] [+/−] [+] [↶] [▸θ,r]

Investigation 1: Find the magnitude r and direction angle ϑ for the vector ⟨12, 5⟩.

Ans: $r = 13$, $\vartheta = 22.62°$

Check Keys: [5] [ENTER] [1] [2] [↶] [▸θ,r]

Investigation 2: Find the hypotenuse of a right triangle with legs of length 7 and 24.

Ans: $r = \sqrt{49 + 576} = 25$

Check Keys: [2] [4] [ENTER] [7] [↶] [▸θ,r]

Investigation 3: In a 30°-60°-90° right triangle, what is sin(30°)?

Ans: 42 / 84 = 0.5

Check Keys: [3] [0] [SIN]

Investigation 4: A vector has magnitude $r = 10$ at angle $\vartheta = 45°$. Find its horizontal and vertical components.

Ans: $x \approx 7.07$, $y \approx 7.07$

Check Keys: [4] [5] [ENTER] [1] [0] [↷] [▸y,x]

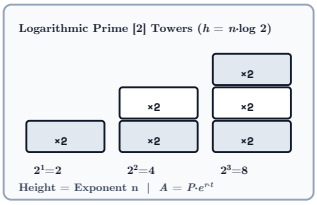
Continuous Compounding & Euler's Constant e

Core Concept Lesson • Big Ideas Math (Exponential & Logarithmic Functions) • Standard: CCSS.MATH.HSF.LE.A.4

1. MATHEMATICAL CONCEPT & STACK APPROACH: Evaluate discrete and continuous exponential growth models from the inside out (exponent first, then base power, then initial value).

Exponential growth $P{\cdot}e^{rt}$ compounds continuously by scaling a base quantity according to an inner rate-time product ($r \times t$). Evaluating from the inside out—computing the exponent r × t first, raising the base, and multiplying by P—follows the exact mathematical hierarchy without parentheses.

Classroom Hook: Stack three [2] Prime Factor blocks (7.0 mm each) beside the 21.0 mm mark for 8 on the comparator scale. How did adding physical block heights multiply 2 × 2 × 2 = 8?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Prime Factor Studio (Logarithmic Blocks) & Growth Curve

Hands-On Setup: Stack Prime Factor [2] blocks (7.0 mm each) to show linear height addition representing exponential doubling ($2^1 = 2$, $2^2 = 4$, $2^3 = 8$).

Predict First (Before Calculator): Invest $1,000 at 7% continuous interest for 10 years: $A = 1000 \times e^{0.07\times10} = 1000 \times e^{0.7}$. Will it double? ($e^{0.7} \approx 2.01$).

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Compute the inner growth exponent $r \times t$ first, press [e^x] to get the growth factor, and then multiply by principal P!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Multiply the base by the exponent or round intermediate decimals to two places."

3. CALCULATOR EXECUTION: SC-32 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [0] [•] [0] [7] [ENTER] [1] [0] [×] [e^x] [1] [0] [0] [0] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[0] [•] [0] [7]	0.07	Rate $r = 0.07$ entered in X
[ENTER]	0.07	Rate lifted to Y
[1] [0]	10	Time $t = 10$ entered in X
[×]	0.7	Exponent $r \times t = 0.70$
[e^x]	2.0138	$e^{0.70} = 2.01375$ (Growth factor: more than doubled!)
[1] [0] [0] [0]	1000	Principal $P = 1000$ (growth factor lifts to Y)
[×]	2013.75	Total accumulated amount $A = \$2{,}013.75$!

4. CONTEXT (John Napier's Logarithms: Turning Multiplication into Height Addition) • Thinking in the Stack (SC-32 Manual) — Ch. 1: 100-Decade Scaling + Ch. 3: Logarithms, Power Laws & Exponential Scaling ([LN], [e^x]) + Ch. 11: Time Value of Money ([TVM])

In 1614, Scottish mathematician John Napier realized that logarithms turn tedious multi-digit multiplication into simple addition—the exact engineering principle behind our Prime Factor Studio blocks, whose physical heights scale as $h = 7.0 \times \log2(p)$ mm so stacking blocks adds their logarithms! Paired with Jakob Bernoulli's continuous compounding constant e, logarithms unlock doubling time and half-life.

Core Investigation (Student Warm-Up A): A bacteria culture starts with $P = 500$ cells and triples every hour. Calculate the population after t = 4 hours.

Worked Answer: $3^4 = 81$; $500 \times 81 = 40{,}500$ bacteria. (All Keys: stackcalc.io/curriculum/answers#unit-10)

Verification Keys: [3] [ENTER] [4] [y^x] [5] [0] [0] [×]

Solving for Time: Doubling Time, Half-Life & Natural Logarithms

Follow-Up Extension (Big Ideas Math • CCSS.MATH.HSF.BF.B.5) + Misconception Remediations & Lab Verification

1. FOLLOW-UP EXTENSION LESSON: CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: A logarithm is the inverse of exponentiation: it converts multiplicative scaling into additive height ($\ln(a \cdot b) = \ln(a) + \ln(b)$). To solve for time t in $P \cdot e^{rt} = A$, divide A by P first, take the natural logarithm $\ln(A/P)$ to unwrap the exponent, and divide by r.

Blocks & Prediction: Observe how Prime Factor block heights scale with ln(p) on the comparator lid while comparing additive block heights to multiplicative products. Prediction: At 5% continuous growth ($r = 0.05$), doubling time is $t = \ln(2) / 0.05 \approx 0.69315 / 0.05 \approx 13.86$ years.

Extension Key Sequence: [2] [LN] [0] [•] [0] [5] [÷]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[2]	2	Doubling target factor 2
[LN]	0.6931	Natural log ln(2) = 0.693147... in X
[0] [•] [0] [5]	0.05	Rate $r = 0.05$ (ln(2) lifts to Y)
[÷]	13.86	$t = \ln(2) / r = 13.86$ years!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Multiplying the base by the exponent (computing 2^5 as $2 \times 5 = 10$) or reversing Base and Exponent on [y^x].

Hands-On Block & Stack Fix: Stack five [2] Prime Factor logarithmic blocks in the comparator lid to show that adding 5 block heights multiplies $2\times2\times2\times2\times2 = 32$, and verify that Base y goes in Register Y and Exponent x goes in X before [y^x].

Misconception 2: Applying the power [y^x] after multiplying by principal P in $P(1 + r)^t$ (computing $(P \cdot (1+r))^t$ instead of $P \cdot ((1+r)^t)$).

Hands-On Block & Stack Fix: Use the Inside-Out RPN rule: compute $(1 + r)^t$ first ([1.05] [ENTER] [2] [y^x]), then multiply by P ([1000] [×]).

3. HANDS-ON LAB COACHING & STACK VERIFICATION KEYS (DOMAIN 10 WARM-UP B & INVESTIGATIONS 1–4)

Warm-Up B (Ext.): A radioactive isotope decays according to $N(t) = N_0 \cdot e^{-0.035t}$. Find its half-life t to two decimal places. **Ans: $t = \ln(2) / 0.035 \approx 0.6931 / 0.035 \approx 19.80$ years.**

Check Keys: [2] [LN] [0] [•] [0] [3] [5] [÷]

Investigation 1: Evaluate e^2 to 4 decimal places. **Ans: 7.3891**

Check Keys: [2] [e^x]

Investigation 2: Find the continuous doubling time t for an investment growing continuously at $r = 8\%$. **Ans: 8.66 years**

Check Keys: [2] [LN] [0] [•] [0] [8] [÷]

Investigation 3: Evaluate $10^{3.5}$ to two decimal places. **Ans: 3162.28**

Check Keys: [3] [•] [5] [↶] [10^x]

Investigation 4: Solve $10^x = 500$ for x to four decimal places. **Ans: 2.6990**

Check Keys: [5] [0] [0] [↶] [LOG]

PART I • DOMAIN 11 • STATISTICS, PROBABILITY & COMBINATORICS • GRADES 11–12 / AP (SC-32 SCIENTIFIC)

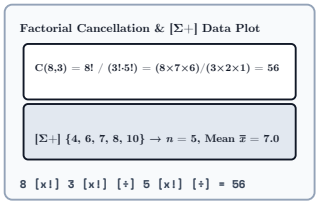
Combinations vs. Permutations: Subsets & Prime Cancellation

Core Concept Lesson • AP Statistics (Probability, Random Variables & Combinatorics) • Standard: CCSS.MATH.HSS.CP.B.9

1. MATHEMATICAL CONCEPT & STACK APPROACH: Calculate permutations and combinations by canceling common factorial terms on paper and verifying on the stack.

Combinations nCr = $n!$ / (r!(n-r)!) count unordered selections by dividing ordered permutations by the r! duplicate arrangements of the chosen group. Canceling 5! from 8! leaves (8 × 7 × 6) / 3! = 56.

Classroom Hook: Why does choosing a 3-person committee from 8 people (8C3 = 56) produce 6 times fewer outcomes than ranking 1st, 2nd, and 3rd place (8P3 = 336)?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Block Kit & Mathematical Focus: Prime Factor Studio (Factorial Cancellation) & Statistical Data Plot

Hands-On Setup: Use Prime Factor blocks to model prime factor cancellation in 8! / (3! × 5!) = (8 × 7 × 6) / (3 × 2 × 1) = 8 × 7 = 56.

Predict First (Before Calculator): How many ways to choose a committee of 3 from 8 candidates? 8C3 = 8! / (3! × 5!) = (8 × 7 × 6) / 6 = 56.

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Press [CLΣ] first, then feed each sample value in with [Σ+]—watch Register X count n while the stack tracks the running mean!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Re-type all 12 numbers from scratch if you want both the mean and standard deviation."

3. CALCULATOR EXECUTION: SC-32 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: 8 ↶ x! 3 ↶ x! ÷ 5 ↶ x! ÷

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
8	8	Total candidates n = 8
↶ x!	40320	8! = 40,320 in register X
3	3	Subset r = 3 (40320 lifts to Y)
↶ x!	6	3! = 6 in X
÷	6720	40320 / 6 = 6720 in X
5	5	Remaining n - r = 5 (6720 lifts to Y)
↶ x!	120	5! = 120 in X
÷	56	6720 / 120 = 56 combinations!

4. CONTEXT (Sir Francis Galton's 1873 Bean Machine & The Mathematics of Small Samples) • Thinking in the Stack (SC-32 Manual) — Ch. 9: Least Squares, Small Samples & Uncertainty (Gauss's Ceres Recovery, Gosset, [PROB], [zt], [nCr], [nPr] & [Σ+])

In 1873, Sir Francis Galton built a vertical board studded with pins (the Quincunx) to show how balls bouncing left or right in combinations nCr settle into a bell-shaped curve—while in 1908 in Dublin, chemist William Sealy Gosset ('Student') developed small-sample statistics with n − 1 degrees of freedom! Students cancel factorial primes with Prime Factor blocks and model exact probability ratios.

Core Investigation (Student Warm-Up A): A lottery draws 4 winning numbers from a pool of 10 (order does not matter). How many winning combinations are possible?

Worked Answer: (10 × 9 × 8 × 7) / (4 × 3 × 2 × 1) = 5040 / 24 = 210 combinations. (All Keys: stackcalc.io/curriculum/answers#unit-11)

Verification Keys: 1 0 ↶ x! 4 ↶ x! ÷ 6 ↶ x! ÷

Sample Mean & Variance of Discrete Data Sets

Follow-Up Extension (AP Statistics • CCSS.MATH.HSS.ID.A.2) + Misconception Remediations & Lab Verification

1. FOLLOW-UP EXTENSION LESSON: CONCEPT → PAPER STEPS → CALCULATOR

Concept & Stack Approach: Statistical measures of center (Mean $\bar{x}$) and spread (Standard Deviation s) summarize an entire distribution of values. Finding the sum of observations (Σx) and dividing by the sample count n locates the balance point of the data set.

Mathematical Setup & Prediction: Plot the 5 sample values {4, 6, 7, 8, 10} on a number-line dot plot to see how distances above and below 7 balance. Prediction: Sum = 4 + 6 + 7 + 8 + 10 = 35. Sample size $n = 5$. Mean $\bar{x} = 35 / 5 = 7.0$.

Extension Key Sequence: [4] [Σ+] [6] [Σ+] [7] [Σ+] [8] [Σ+] [1] [0] [Σ+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[4] [Σ+]	1	First score entered: sample size $n = 1$
[6] [Σ+]	2	Second score: $n = 2$
[7] [Σ+]	3	Third score: $n = 3$
[8] [Σ+]	4	Fourth score: $n = 4$
[1] [0] [Σ+]	5	Fifth score: $n = 5$ total data points

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Forgetting to clear the statistical registers ([CLΣ]) before entering a new dataset with [Σ+].

Hands-On Block & Stack Fix: Notice that pressing [Σ+] displays the sample count n in Register X; if the first data point shows $n > 1$, press [CLΣ] first so prior data does not corrupt the mean.

Misconception 2: Dividing by r! and then multiplying by $(n - r)!$ when computing C(n, r) = $n!$ / (r!$(n - r)$!).

Hands-On Block & Stack Fix: Compute n! into Y, compute r! and $(n - r)$! and multiply them together in X ([×]), then press [÷] once to divide n! by the combined denominator.

3. HANDS-ON LAB COACHING & STACK VERIFICATION KEYS (DOMAIN 11 WARM-UP B & INVESTIGATIONS 1–4)

Warm-Up B (Ext.): Given the four data points {12, 15, 18, 21}, find the sample size n, the total sum of the values, and the sample mean $\bar{x}$.
Check Keys: [1] [2] [Σ+] [1] [5] [Σ+] [1] [8] [Σ+] [2] [1] [Σ+]
Ans: Count $n = 4$, Sum = 66, Mean $\bar{x} = 66 / 4 = 16.5$.

Investigation 1: Evaluate 6! (the number of ways to arrange 6 distinct books on a shelf).
Check Keys: [6] [↶] [x!]
Ans: $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

Investigation 2: How many ways can 5 runners finish in 1st, 2nd, and 3rd place?
Check Keys: [5] [↶] [x!] [2] [↶] [x!] [÷]
Ans: $5 \times 4 \times 3 = 60$

Investigation 3: How many 2-person partnerships can be formed from 7 students?
Check Keys: [7] [↶] [x!] [2] [↶] [x!] [÷] [5] [↶] [x!] [÷]
Ans: $(7 \times 6) / 2 = 21$

Investigation 4: Find the mean $\bar{x}$ of the two data values 5 and 15.
Check Keys: [5] [Σ+] [1] [5] [Σ+] [→] [$\bar{x}$,$\bar{y}$]
Ans: $(5 + 15) / 2 = 10$

PART I • DOMAIN 12 • DIGITAL LOGIC, BASE-N & COMPUTER ARCHITECTURE • GRADES 9–12 / AP CS (SC-32 SCIENTIFIC)

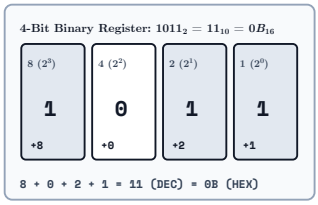
Positional Number Systems: Decimal, Binary & Hexadecimal

Core Concept Lesson • AP Computer Science Principles (Data Representation & Binary Number Systems) • Standard: CCSS.MATH.PRACTICE.MP5

1. MATHEMATICAL CONCEPT & STACK APPROACH: Decompose integers into powers of 2 and powers of 16 using [0] and [1] blocks on the Place Value Tray, and verify base conversions.

Binary (base-2) and hexadecimal (base-16) follow the exact same positional place-value principle as base-10, using powers of 2 (8, 4, 2, 1) or 16 instead of powers of 10. Every 4-bit binary group (0000 to 1111) maps to exactly one hexadecimal digit (0 to F).

Classroom Hook: How can you represent every number from 0 to 15 using only [0] and [1] Digit Blocks across 4 sockets of your Place Value Tray?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Binary [0]/[1] Register Trays & Base-N Columns)

Hands-On Setup: Use the sockets of the Place Value Tray as binary bit positions (8, 4, 2, 1). Place [1] Digit Blocks for active '1' bits and [0] Digit Blocks for '0' bits.

Predict First (Before Calculator): Place [1], [0], [1], [1] in the 8, 4, 2, 1 sockets: binary $1011_2 = 8 + 0 + 2 + 1 = 11_{10}$. In hexadecimal ($A$=10, B=11), $11_{10} = 0B_{16}$.

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Place [1] and [0] Digit Blocks in the 8-4-2-1 sockets of your Place Value Tray, then toggle [BIN] and [HEX] on the SC-32 to see the exact same bit pattern!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Convert binary to decimal only by long-division remainders on scratch paper."

3. CALCULATOR EXECUTION: SC-32 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [1] [1] [↶] [BASE]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[1] [1]	11	Enter decimal integer 11 in DEC mode
[↶] [BASE]	HEX 0B	Press [f:] [BASE]: screen displays Hexadecimal 0B!
[↶] [BASE]	BIN 1011	Press [f:] [BASE]: screen displays Binary 1011!
[↶] [BASE]	OCT 13	Press [f:] [BASE]: screen displays Octal 13!
[↶] [BASE]	DEC 11	Press [f:] [BASE]: returns to Decimal 11

4. CONTEXT (Leibniz's Binary Medallion & Claude Shannon's Boolean Relay Circuits) • Thinking in the Stack (SC-32 Manual) — Ch. 1: 32,768 Hz Quartz Timebase + Ch. 10: Boolean Logic, Two's Complement & the 36-Bit Word ([BASE]) + Ch. 13: Coupled Systems

In 1697, Gottfried Wilhelm Leibniz designed a medallion showing that every integer can be built from 1 and 0 alone using powers of 2 (8, 4, 2, 1)—and in 1937, 21-year-old MIT student Claude Shannon proved that electrical switches wired in series (AND) and parallel (OR) execute Boolean logic directly! Using [0] and [1] blocks on the Place Value Tray, students see how place value powers modern computing.

Core Investigation (Student Warm-Up A): Express decimal 255 as a sum of powers of 2 to write its 8-bit binary and 2-digit hexadecimal representations.

Worked Answer: Binary: 11111111_2 (eight 1*s*); Hexadecimal: FF_{16} ($15 \times 16 + 15 = 255$). (All Keys: stackcalc.io/curriculum/answers#unit-12)

Verification Keys: [2] [5] [5] [↶] [BASE]

Bitwise Logic: AND Masking & Column-by-Column Comparison

Follow-Up Extension (AP Computer Science Principles • CCSS.MATH.PRACTICE.MP5) + Misconception Remediations & Lab Verification

1. FOLLOW-UP EXTENSION LESSON: CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Bitwise logic operations (AND, OR, XOR) compare corresponding binary place-value columns of two integers in parallel. Placing value A in the top row and mask B in the bottom row lets you inspect each bit column directly.

Blocks & Prediction: Row 1 of the Place Value Tray holds nibble A ($[1][1][0][0]_2 = 12$). Row 0 holds mask B ($[1][0][1][0]_2 = 10$). Prediction: Bitwise AND: bit 3 (1 & 1 = 1), bit 2 (1 & 0 = 0), bit 1 (0 & 1 = 0), bit 0 (0 & 0 = 0). Result: $1000_2 = 8_{10}$.

Extension Key Sequence: [1] [2] [ENTER] [1] [0] [AND]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1] [2]	12	Enter operand $A = 12$ (binary 1100)
[ENTER]	12	12 lifted to register Y
[1] [0]	10	Enter operand $B = 10$ (binary 1010) in X
[AND]	8	Bitwise AND evaluates 1100 & 1010 = 1000 (8 decimal)!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Confusing bitwise [AND] / [OR] with decimal addition or multiplication (e.g., thinking 12 AND 10 is 22).

Hands-On Block & Stack Fix: Model 12 ($[1][1][0][0]_2$) in Row 1 and 10 ($[1][0][1][0]_2$) in Row 0 of the Place Value Tray using [1] and [0] Digit Blocks; only columns where BOTH rows hold [1] remain 1 in 12 AND 10 = 8 ($[1][0][0][0]_2$).

Misconception 2: Reading hexadecimal [10] as ten instead of decimal 16 ($1 \times 16^1 + 0 \times 16^0$).

Hands-On Block & Stack Fix: Place [1] in the Sixteens column and [0] in the Ones column on the Place Value Tray, then switch from [HEX] [10] to [DEC] on the SC-32 to see 16.

3. HANDS-ON LAB COACHING & STACK VERIFICATION KEYS (DOMAIN 12 WARM-UP B & INVESTIGATIONS 1–4)

Warm-Up B (Ext.): Compare 12 and 3 column by column in 4-bit binary to find the result of 12 OR 3 in both binary and decimal.
Ans: 1100_2 OR $0011_2 = 1111_2 = 15_{10}$.
Check Keys: [1] [2] [ENTER] [3] [OR]

Investigation 1: Decompose decimal 42 into powers of 2 to write its 6-bit binary representation.
Ans: $32 + 8 + 2 \rightarrow 101010_2$
Check Keys: [4] [2] [↶] [BASE]

Investigation 2: Convert hexadecimal $2A_{16}$ to decimal using place value.
Ans: $32 + 10 = 42$
Check Keys: [2] [ENTER] [1] [6] [×] [1] [0] [+] [↶] [BASE]

Investigation 3: Evaluate the bitwise AND of 15 and 7 in binary and decimal.
Ans: $0111_2 = 7$
Check Keys: [1] [5] [ENTER] [7] [AND]

Investigation 4: Evaluate the bitwise XOR of 12 and 10 in binary and decimal.
Ans: $0110_2 = 6$
Check Keys: [1] [2] [ENTER] [1] [0] [XOR]

Middle School Section & Concept Coaching

Teacher coaching notes, misconception alerts, and two-method facilitation for Book 3 (Middle School Math Lab).

HOW PART II CONNECTS CONCEPT DOMAINS TO DAILY MIDDLE SCHOOL SECTIONS

While Part I organizes K–12 instruction by macro Concept Domain, middle school teachers using Book 3 (Middle School Math Lab — Aligned to Big Ideas Math Grades 6–8) also need rapid unit-by-unit coaching notes connecting each chapter's Concept Approach, Physical Block Models, and SC-6 Stack Verification strategies. Pages 29–32 provide the 10-Unit Middle School Concept-Blocks-Calculator Coaching Guide, Misconception Matrix, and Classroom Mastery Roster (with complete worked keys at stackcalc.io/curriculum/answers).

Middle School Unit (Book 3)	Sections & Core Concepts	Related Part I Domains	Coach Page
Unit 01: Numerical Expressions & Factors	Sec 1.1–1.5 (Powers, Order of Ops, Prime Factor, GCF/LCM)	**Domains 05 & 07**	**p. 29**
Unit 02: Fractions & Decimals	Sec 2.1–2.5 (Multiplying/Dividing Fractions & Decimals)	**Domains 05 & 06**	**p. 29**
Unit 03: Ratios, Rates & Percents	Sec 3.1–3.5 (Ratios, Tape Diagrams, Unit Rates & Percents)	**Domains 04 & 06**	**p. 29**
Unit 04: Algebraic Expressions & Properties	Sec 4.1–4.4 (Evaluating, Writing, Properties & Factoring)	**Domains 07 & 08**	**p. 29**
Unit 05: Areas of Polygons & Elapsed Time	Sec 5.1–5.4 (Parallelograms, Triangles, Trapezoids & Time)	**Domains 05 & 09**	**p. 29**
Unit 06: Integers, Rationals & Coord. Plane	Sec 6.1–6.5 (Signed Numbers, Absolute Value & 4 Quadrants)	**Domains 03, 06 & 08**	**p. 30**
Unit 07: Equations & Inequalities	Sec 7.1–7.5 (1-Step Equations, Tables/Graphs & Inequalities)	**Domains 07 & 08**	**p. 30**
Unit 08: Surface Area & Volume of 3D Solids	Sec 8.1–8.4 (3D Nets, Prisms, Pyramids & Fractional Edge Vol)	**Domains 06 & 09**	**p. 30**
Unit 09: Statistical Measures & Data Displays	Sec 9.1–9.4 (Mean, Median, Mode, IQR, MAD & Box Plots)	**Domain 11**	**p. 30**
Unit 10: Grade 7 Bridge — Signed, 2-Step & Prob.	Sec 10.1–10.4 (Signed Operations, 2-Step Eq & Probability)	**Domains 08, 09 & 11**	**p. 30**
Middle School Misconception Matrix	10 High-Frequency Errors & Block Remediations	**All Domains**	**p. 31**
Section Mastery & Token Roster	45-Section Pacing & Classroom Check-Off Grid	**All Domains**	**p. 32**

Dual-Method Paper-First Mastery & Online Solution Portal (stackcalc.io/curriculum/answers)

Every problem in Book 3 requires (1) written paper setup first, followed by (2) stack verification on the SC-6. Pages 29–30 summarize the Concept → Blocks → Calculator triad for Units 01–10; complete worked answers are hosted at https://stackcalc.io/curriculum/answers.

Middle School Units 01–05 Coaching Guide

Units 01 to 05 (Expressions/Factors, Fractions/Decimals, Ratios/Percents, Algebra & Polygon Area)

Middle School Unit 01: Numerical Expressions & Factors • Sections 1.1–1.5 (5 sections • 100 problems)

1. Concept (Stack Approach): Multi-step expressions evaluate from the inside out—holding completed groups on shelf Y while building the next group on X. Every whole number factors uniquely into primes.

2. Blocks (Place Value & Prime): Model division quotients/remainders on the Place Value tray and stack Prime Factor blocks to see exponents (2^3), GCF (overlap), and LCM (union).

3. Calculator (SC-6): Use [g][÷R] to split Remainder (Y) and Quotient (X), [g][FACT] for prime powers, and [GCF]/[f][LCM] to verify Venn diagram results.

Middle School Unit 02: Fractions & Decimals • Sections 2.1–2.5 (5 sections • 100 problems)

1. Concept (Stack Approach): Rational operations require common units before adding/subtracting and multiplicative inverses (reciprocals) when dividing, holding the first operand on Y while transforming X.

2. Blocks (Fraction & Place Value): Compare 84 mm Fraction Strips side-by-side to see common denominators and 'how many fit' division; align tenths and hundredths on the Decimal Tray.

3. Calculator (SC-6): Enter fractions with Two-Dot notation ([.] n [.] d), flip only the divisor in X with [f][1/x] before [×], and split decimals with [f][IP] and [g][FP].

Middle School Unit 03: Ratios, Rates & Percents • Sections 3.1–3.5 (5 sections • 100 problems)

1. Concept (Stack Approach): Ratios, unit rates, and percents are multiplicative comparisons. Keeping Y and X paired preserves the two quantities being compared across scaling steps.

2. Blocks (Fraction & Place Value): Lay out equal-length Fraction Strips as physical Tape Diagram bars to distinguish Part-to-Part (3:2) from Part-to-Whole (3/5) and model benchmark percents.

3. Calculator (SC-6): Simplify ratios on Y:X with [g][a:b], step through ratio tables with [g][REP], swap upside-down unit rates with [$x\rightleftarrows y$], and toggle forms with [FDISP].

Middle School Unit 04: Algebraic Expressions & Properties • Sections 4.1–4.4 (4 sections • 80 problems)

1. Concept (Stack Approach): Algebraic expressions are recipes of operations waiting for an input value. Evaluating an expression means pushing the substituted value onto the stack and executing inner terms first.

2. Blocks (Place Value & Prime): Treat 1, x, and x^2 as distinct Place Value columns so unlike terms are never combined, and use Prime Factor blocks to factor out the GCF.

3. Calculator (SC-6): Substitute a test value (e.g., $x = 5$) into both original and simplified expressions on Y and X, then press [≠] (flashes NO when equivalent).

Middle School Unit 05: Areas of Polygons & Elapsed Time • Sections 5.1–5.4 (4 sections • 80 problems)

1. Concept (Stack Approach): Every polygon area formula decomposes into base-times-height products held on the stack, while elapsed time regroups in base-60 rather than base-10.

2. Blocks (When Applicable — Fraction & Place Value): Use Fraction Strips for fractional edge lengths and Place Value columns to contrast base-10 regrouping with 60-minute hours.

3. Calculator (SC-6): Compute rectangle [f][RECT] and triangle [g][TRI] areas, accumulate composite sub-areas with [Σ+] and [f][Σx], and convert H.MMSS time via [f][→HR] and [g][→HMS].

Middle School Units 06–10 Coaching Guide

Units 06 to 10 (Integers/Coord. Plane, Equations/Inequalities, 3D Solids, Statistics & Grade 7 Bridge)

Middle School Unit 06: Integers, Rational Numbers & The Coordinate Plane • Sections 6.1–6.5 (5 sections • 100 problems)

1. Concept (Stack Approach): A signed number combines direction (+/-) and magnitude (|x|). In postfix entry, typing the magnitude first and pressing [+/-] attaches the sign to the number in X.

2. Blocks (Place Value & Fraction): Pair positive and negative Place Value blocks to form Zero Pairs, and compare rational magnitudes with 84 mm Fraction Strips.

3. Calculator (SC-6): Distinguish [+/-] (change sign of X) from [-] (subtract X from Y), animate signed hops on [f][LINE], find distance with [g]|x|, and test order with [f][>]/[g][<].

Middle School Unit 07: Equations & Inequalities • Sections 7.1–7.5 (5 sections • 100 problems)

1. Concept (Stack Approach): An equation is a balanced two-register comparison ($Y = X$). Solving isolates the unknown by applying inverse operations in reverse order.

2. Blocks (Place Value & Fraction): Model both sides of an equation on the Place Value tray or with Fraction Strips so removing/dividing equal amounts preserves balance.

3. Calculator (SC-6): Solve with inverse operations on the stack, then push the left and right sides into Y and X and press [≠] (flashes NO when both sides are equal).

Middle School Unit 08: Surface Area & Volume of 3D Solids • Sections 8.1–8.4 (4 sections • 80 problems)

1. Concept (Stack Approach): Surface area sums 2D face areas across an unfolded net, whereas 3D volume multiplies the 2D base area B (in Y) by the perpendicular height h (in X).

2. Blocks (When Applicable — Place Value & Fraction): Pack Place Value unit blocks to see $V = B{\cdot}h$ in layers, and use Fraction Strips to model fractional edge lengths.

3. Calculator (SC-6): Accumulate opposite face pairs with [Σ+] → [f][Σx] for surface area, multiply Base Area × Height with [f][RECT] → [f][PRISM] for volume, and verify [f][CIRC] → [g][CONE] and [g][HYP].

Middle School Unit 09: Statistical Measures & Data Displays • Sections 9.1–9.4 (4 sections • 80 problems)

1. Concept (Stack Approach): Statistical summaries condense a list of numbers into measures of center (Mean, Median) and spread (IQR, MAD) by accumulating data into dedicated statistical registers.

2. Blocks (Place Value & Fraction): Level unequal Place Value digit stacks to model the Mean as a fair share, and use 4 equal Fraction Strips to visualize 25% quartiles.

3. Calculator (SC-6): Clear with [g][CLΣ], enter data once with [Σ+] (or [FRQ]), and read [g][$\bar{x}$], [f][MED], [g][MODE], [RNG], [f][1Q], [g][3Q], [IQR], and [g][MAD].

Middle School Unit 10: Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability • Sections 10.1–10.4 (4 sections • 80 problems)

1. Concept (Stack Approach): Two-step equations unwrap in reverse order of operations (undo $\pm b$ first, then $\div a$ second), and squaring a negative in X ([+/-][x^2]) squares the entire signed value.

2. Blocks (Place Value, Fraction & Prime): Model zero pairs and 2-step balance with Place Value blocks and probability part-to-whole ratios with Fraction Strips.

3. Calculator (SC-6): Verify signed powers (5 [+/-][x^2] = +25), check 2-step equations with [≠], and evaluate factorials [$x!$], combinations [f][nCr], and permutations [g][nPr].

Middle School 10-Unit Misconception & Hands-On Remediation Matrix

UNIT & HANDS-ON MODEL	TARGET CLASSROOM MISCONCEPTION	HANDS-ON BLOCK & SC-6 REMEDIATION
Unit 1: Numerical Expressions & Factors Place Value & Prime Factor Blocks	Students interpret remainders as meaningless decimal tails (e.g., .875 of a person), read 2^3 as 2×3=6, lose track of PEMDAS parentheses, and confuse GCF (overlap) with LCM (union).	Have students model division remainders on the Place Value Tray, stack Prime Factor Blocks ([2][2][2] = 2^3) for exponents, use the 4-line elevator (Y holds Room 1 while X solves Room 2) instead of rewriting PEMDAS lines, and stand logarithmic Prime Factor Towers side-by-side on the 12-socket lid.
Unit 2: Fractions & Decimals Fraction Strips & Place Value Blocks	Students treat fractions, mixed numbers, and decimals as unrelated rules, expect multiplication to always make numbers bigger, forget which fraction to flip in division, and misalign decimal columns.	Use the 84.0 mm Fraction Strips towers so equivalence and common denominators are physical millimeter height matches. Use [f] [1/x] on Floor 1 (X) so students see only the divisor flip, and use [f] [IP] / [g] [FP] to split whole dollars from decimal cents.
Unit 3: Ratios, Rates & Percents Fraction Strips & Place Value Blocks	Students confuse Part-to-Part (3:2) with Part-to-Whole (3/5), add instead of multiply in ratio tables, divide unit rates upside-down, and memorize 6 disconnected rules for moving decimal points in percents.	Use equal Fraction Strips blocks as physical Tape Diagram units, step through ratio tables rhythmically with [g] [REP], use [x swap y] when unit-rate division is entered upside-down, and use [f] [→%] + [FDISP] to show Fraction/Decimal/Percent are three costumes of one number.
Unit 4: Algebraic Expressions & Properties Place Value & Prime Factor Blocks	Students panic when letters appear, erroneously combine $3x + 2$ into $5x$ or $x^2 + x$ into x^3, and don't see why $3(x + 4) = 3x + 12$.	Show that a variable x is simply an empty Place Value tray socket waiting for a Digit Block to be substituted in, and that different place-value columns (x^2, x, and 1s) cannot be combined into one column! Use Prime Factor Towers to find the GCF when factoring expressions.
Unit 5: Areas of Polygons & Elapsed Time Fraction Strips & Place Value Blocks	Students multiply the slanted side of a parallelogram instead of the vertical height, forget to divide by 2 for triangles, lose track of multi-piece composite areas, and subtract clock times in base-10 instead of base-60.	Slide the corner triangle of a parallelogram to form a straight [f] [RECT] rectangle, use the dedicated [g] [TRI] key which divides by 2 automatically, accumulate composite chunks with [Σ+], and use the Place Value Blocks + [CLK] key for base-60 time.
Unit 6: Integers, Rational Numbers & The Coordinate Plane Place Value Blocks & Fraction Strips	Students confuse the subtraction operator [-] with the negative sign [+/-], think -18 > -5 because 18 > 5, confuse Opposite ([+/-]) with Absolute Value ([g] [|x|]), and reverse (x, y) coordinates.	Use Place Value Blocks tokens and the SC-6 Number Line ([f] [LINE]) to model opposites and zero pairs, test inequalities with [f] [>] and [g] [<], contrast [+/-] (flip sign) with [g] [|x|] (always positive distance from 0), and use the Elevator Rule (walk across x first, ride up/down y second).
Unit 7: Equations & Inequalities Place Value Blocks & Fraction Strips	Students treat '=' as 'write the answer', translate '8 less than n' backwards as 8 - n, resent writing -14 under both sides of $x + 14 = 39$, and confuse open vs. solid inequality circles.	Model '=' as a Balance Scale using Place Value Blocks and Fraction Strips. Solve 1-step equations with inverse operations on both sides, verify your solution with the SC-6 Equality Test [≠] (which flashes NO when Left = Right!), and test inequalities on [f] [LINE] with [f] [≥] and [g] [≤] (which flash YES or NO).
Unit 8: Surface Area & Volume of 3D Solids Place Value Blocks & Fraction Strips	Students drop one of the 6 faces when finding rectangular prism surface area, tangle prism and pyramid formulas, and confuse 2D Surface Area (cardboard outside) with 3D Volume (cubes inside).	Have students count the 3 pairs of opposite faces on a rectangular prism while logging each pair into [Σ+] (n=1, 2, 3), pack unit blocks or use [f] [PRISM] to multiply 3 stack floors (l, w, h) at once for volume, and verify [f] [CIRC], [g] [CONE], and [g] [HYP].
Unit 9: Statistical Measures & Data Displays Place Value Blocks & Fraction Strips	Students forget to sort data before finding Median/Quartiles, lose their place adding long lists for Mean, and dread Mean Absolute Deviation (MAD) because doing it by hand takes 20 minutes of subtraction.	Show that Mean is physically leveling unequal Place Value digit stacks to the same height. Then show that entering a list ONCE with [Σ+] unlocks Mean ([g] [x-bar]), Median ([f] [MED]), Mode ([g] [MODE]), Range ([RNG]), Quartiles ([f] [1Q], [g] [3Q], [IQR]), and MAD ([g] [MAD]) in 1 click each!
Unit 10: Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability Place Value, Fraction & Prime Blocks	Students guess signs when adding/subtracting negatives, get -25 instead of +25 for $(-5)^2$ on algebraic calculators, undo ×3 before +7 in $3x + 7 = 28$, and confuse Permutations (order matters) with Combinations (order doesn't).	Use Place Value blocks and the SC-6 Number Line ([f] [LINE]) to show that removing a negative debt (10 - (-4)) equals +14. Notice how RPN 5 [+/-] [x^2] squares the entire register $(-5)^2 = +25$ without parentheses! Unwrap 2-step equations in reverse (undo ±b first, then ÷a second), and finish with [DICE], [f] [nCr], [g] [nPr], and [f] [?].

Photocopiable 10-Unit Middle School Student Mastery Tracker

STUDENT NAME / PERIOD	U1	U2	U3	U4	U5	U6	U7	U8	U9	U10
01. _____										
02. _____										
03. _____										
04. _____										
05. _____										
06. _____										
07. _____										
08. _____										
09. _____										
10. _____										
11. _____										
12. _____										
13. _____										
14. _____										
15. _____										
16. _____										
17. _____										
18. _____										
19. _____										
20. _____										

TEACHER OBSERVATION NOTES & INTERVENTION LOG

- **Print & Digital Student Workbooks & Teacher Guides (The StackCalc Library): https://stackcalc.io/curriculum/**
- Companion Book — From Blocks to the Stack (Amazon): https://link.amazon/B0f5cP0Ur | Manipulatives: https://stackcalc.io

Online Answer Keys & Stack Verification

Why printed answer keys are omitted from the workbooks—and how to access all worked solutions at stackcalc.io/curriculum/answers.

1. WHY WE OMIT PRINTED ANSWER KEYS FROM STUDENT WORKBOOKS (PAPER FIRST → STACK CHECK)

When a student workbook prints an answer key in the back, learners are tempted to flip to the back page the moment a problem feels challenging—short-circuiting productive struggle. By omitting printed answer tables from the workbooks, we establish a healthier two-step loop:

- Step 1 — Solve on Paper & Verify on the Stack: After solving a problem by hand on paper, the student keys their expression into the SC-6 or SC-32 stack. Because the 4-line stack has no [=] or parentheses keys, checking on the stack requires genuine structural understanding.
- Step 2 — Complete Online Worked Solutions & Keystroke Tables (stackcalc.io/curriculum/answers): Teachers, homeschool parents, and coaches have instant free access to every worked answer and canonical keystroke sequence online.

2. COMPLETE ONLINE ANSWER KEY DIRECTORY (HTTPS://STACKCALC.IO/CURRICULUM/ANSWERS)

Workbook Volume & Grade Band	Units, Worksheets & Assessments Covered	Direct Online Answer Key URL
Book 2A: Early Elementary Math Lab	Grades K–2 (SC-6) • Units 01–10: Warm-Ups A/B, Inv. 1–4, Tier 1–3 Practice A–F, Quizzes, STEM & 4 Cumulative Exams (840 problems)	`stackcalc.io/curriculum/answers#early-elementary`
Book 2B: Upper Elementary Math Lab	Grades 3–5 (SC-6) • Units 01–10: Warm-Ups A/B, Inv. 1–4, Tier 1–3 Practice A–F, Quizzes, STEM & 4 Cumulative Exams (840 problems)	`stackcalc.io/curriculum/answers#upper-elementary`
Book 3: Middle School Math Lab	Grades 6–8 (SC-6) • Units 01–10 (45 Sections): Practice A & B (900 problems) + 10 End-of-Unit Mastery Quizzes	`stackcalc.io/curriculum/answers#middle-school`
Book 4: High School & STEM Lab	Grades 9–12 & AP (SC-32) • Units 01–10: Warm-Ups A/B, Inv. 1–4, Tier 1–3 Practice A–F, Quizzes, STEM & 4 Cumulative Exams (840 problems)	`stackcalc.io/curriculum/answers#high-school`
Book 5: Stack Speed & Fluency	Full-Year Edition • All 15 Sections: Warm-Ups W1–W6, Champion Pipelines & 180 Timed Tests A1–D3 (3,690 problems)	`stackcalc.io/curriculum/answers#speed-fluency`
Standalone Printable Unit Packets	Units 01–12 Individual Chapter Packets: Core & Extension Warm-Ups, Lab Investigations 1–4 & Practice Worksheet A	`stackcalc.io/curriculum/answers#unit-packets`

3. AUTOMATED UNIT-TEST VERIFICATION GUARANTEE (EVERY KEYSTROKE & ANSWER 100% TESTED)

Every problem, worked example, warm-up drill, and keystroke sequence across all StackCalc workbooks is continuously verified by automated unit tests against the exact RPNCore 4-register stack simulator and Swift CalculatorEngine. Teachers and students can also test any keystroke sequence interactively in the free browser simulator at https://stackcalc.io.

PYTHON RPN ENGINE SUITE	SWIFT RPNCORE XCTEST SUITE	WEB SIMULATOR PARITY
19 Automated Python Suites	WorkbookCurriculumVerificationTests	https://stackcalc.io (100% Match)

Sample SC-6 Verification (Unit 06 Fractions: $1/3 + 1/4 = 7/12$):



Sample SC-32 Verification (Unit 08 Quadratics: $x^2 + 5x + 6 = 0$):



Manipulative Kits & Keycap Quick-Reference

How each physical manipulative models the 4-register RPN stack, plus essential SC-6 and SC-32 keycap sequences.

Classroom Manipulative Kit	Target Domains	Core Mathematical & RPN Stack Connection
Block Kit 1 • Place Value Studio: Digit Blocks (0–9)	**Domains 01, 02, 04**	Subitizing, part-part-whole bonds (Y + X) & doubles ([ENTER][+])
Block Kit 1 • Place Value Studio: [+1] Carry & Trays	**Domains 03, 05**	10:1 column overflow trade, 6×2 Whole/Decimal alignment & [PV]
Block Kit 1 • Place Value Studio: Binary Bit Sockets	**Domain 12**	0 & 1 bit blocks in powers-of-2 columns (128..1) & bitwise logic
Block Kit 2 • Fraction Studio: 84 mm Fraction Towers	**Domain 06**	1 Whole = 84 mm down to 1/12; two-dot fraction entry ([.][n][.][d])
Block Kit 2 • Fraction Studio: 12-Socket Comparator	**Domain 06 • Book 2B**	Flush-top equivalence check, common denominators & mixed numbers
Block Kit 2 • Fraction Studio: 4-Face Engraved Blocks	**Domain 06 • Book 3**	Rotate 90° to read Fraction (1/4), Decimal (0.25), Percent (25%) & Key (.1.4)
Block Kit 3 • Prime Factor Studio: Logarithmic Blocks	**Domains 07, 10**	Heights $h(p) = 7.0 \cdot \log_2(p)$ mm ([2] = 7 mm = 1 U); stacking adds logs
Block Kit 3 • Prime Factor Studio: GCF/LCM Comparator	**Domains 07, 08, 10, 11**	Shared prime sub-tower = [GCF]; union of prime towers = [LCM]
Instrument 1 • StackCalc SC-6 Elementary Calculator	**Domains 01–07 • K–8**	4-Register Visible Stack (T, Z, Y, X), Two-Dot Fractions (.a.b), [PV], [FACT], [GCF]
Instrument 2 • StackCalc SC-32 Scientific Calculator	**Domains 08–12 • 9–12**	32-Key Scientific RPN: [POLY], [$\rightarrow\vartheta,r$], [SIN/COS/TAN], [LN/LOG], [nCr], [BASE]

CANONICAL KEYCAP SEQUENCES • CLASSROOM [FLAGS]: [6] SILENT • [7] 2/4-STACK • [8] MANUAL [SIMP]

Chained Expression (3×4)+(5×6) = 42	**SC-6**	3 ENTER 4 × 5 ENTER 6 × +
Two-Dot Fraction 2/3 × 3/4 = 6/12 → [SIMP] 1/2	**SC-6**	• 2 • 3 ENTER • 3 • 4 × g SIMP
Mixed Number 2 3/8 → Improper 19/8 ([d/c])	**SC-6**	2 • 3 • 8 ENTER f d/c
Quotient/Remainder [÷R] & Inverse Check [LASTX]	**SC-6**	2 3 ENTER 4 g ÷R f LASTx
Prime Factorization & GCF(24, 36) = 12	**SC-6**	2 4 g 人 2 4 ENTER 3 6 GCF
Quadratic Roots of $x^2 + 5x + 6 = 0$	**SC-32**	↗ POLY 1 B 5 C ON 6 D XEQ
2D Polar Vector (x=3, y=4) → r=5, ϑ=53.13°	**SC-32**	4 ENTER 3 ↖ ▸θ,r
Continuous Growth 500 × $e^{0.04\times5}$	**SC-32**	• 0 4 ENTER 5 × e^x 5 0 0 ×

Concept Pacing & Classroom Observation Log

Photocopiable yearly planning tracker for Concept Domains 01–12 and Middle School Units 01–10.

Concept Domain & Topic	Dates Taught	Components Completed	Student Mastery Notes
Domain 01: Cardinality, Counting & Step Accumulation	_____/_____/20__	[ ]Labs [ ]A–B [ ]C–D [ ]E–F/Quiz	
Domain 02: Part-Part-Whole Number Bonds & Missing Addends	_____/_____/20__	[ ]Labs [ ]A–B [ ]C–D [ ]E–F/Quiz	
Domain 03: Complements to 10 & Gold [+1] Carry Regrouping	_____/_____/20__	[ ]Labs [ ]A–B [ ]C–D [ ]E–F/Quiz	
Domain 04: Twin-Row Place Value Tray: Doubles & Equal Groups	_____/_____/20__	[ ]Labs [ ]A–B [ ]C–D [ ]E–F/Quiz	
Domain 05: Place Value & The 4-Level Stack Elevator	_____/_____/20__	[ ]Labs [ ]A–B [ ]C–D [ ]E–F/Quiz	
Domain 06: Fraction Studio: 84 mm Towers & Two-Dot Entry	_____/_____/20__	[ ]Labs [ ]A–B [ ]C–D [ ]E–F/Quiz	
Domain 07: Postfix Trees, Order of Operations & Prime Factors	_____/_____/20__	[ ]Labs [ ]A–B [ ]C–D [ ]E–F/Quiz	
Domain 08: Base-x Polynomials, Prime Factor Pairs & Roots	_____/_____/20__	[ ]Labs [ ]A–B [ ]C–D [ ]E–F/Quiz	
Domain 09: Coordinate Geometry, Trigonometry & Vectors	_____/_____/20__	[ ]Labs [ ]A–B [ ]C–D [ ]E–F/Quiz	
Domain 10: Exponents, Logarithms & Continuous Growth	_____/_____/20__	[ ]Labs [ ]A–B [ ]C–D [ ]E–F/Quiz	
Domain 11: Combinatorics, Probability Ratios & Series Statistics	_____/_____/20__	[ ]Labs [ ]A–B [ ]C–D [ ]E–F/Quiz	
Domain 12: Digital Logic, Base-N & Computer Architecture	_____/_____/20__	[ ]Labs [ ]A–B [ ]C–D [ ]E–F/Quiz	

5-DAY PACING, 3-TIER DIFFERENTIATION (PRACTICE A–F) & CLASSROOM PROTOCOLS

5-Day Unit Rhythm: Day 1 Core/Ext & Inv • Day 2 Tier 1 (A–B Foundation) • Day 3 Tier 2 (C–D Core) • Day 4 Tier 3 (E–F Challenge) • Day 5 Quiz & STEM

Protocols: Partner A (Architect: Blocks +10) & Partner B (Navigator: Stack +10) • Swap every 3 probs • Hand-Signals: *X*=Desk, *Y*=Chest, *Z*=Shoulder, *T*=Overhead

Homeschool & Family Guide to Stack Mathematics

How tactile blocks, paper-first mastery, and a distraction-free stack help children flourish at home and in class.

DEAR PARENTS & FAMILIES: WHY WE PAIR TACTILE BLOCKS & PAPER MATH WITH A 4-LINE STACK

If you value strong mental math, paper-and-pencil discipline, and a screen-free home, you might wonder why our program includes a calculator—and why it has an [ENTER] key instead of an [=] sign! Here is why this system helps students flourish:

• Zero Screens, Apps, or Crutches: The SC-6 and SC-32 are quiet, single-purpose instruments (like an abacus or slide rule—no Wi-Fi, games, or notifications). Students NEVER use the calculator to skip paper work: they build with blocks and solve on paper FIRST.

• Mirrors Paper-and-Pencil Math: On paper, you write the first number on the top line (Line Y), write the second number below it (Line X), and then add (+). Because our 4-line screen has no parentheses keys, students must solve from the inside out—verifying their own paper work independently at the kitchen table without peeking at a back-of-book answer key!

THE 3-EXAMPLE KITCHEN-TABLE CHEAT SHEET (TRY THIS WITH YOUR STUDENT IN 2 MINUTES!)

Example 1: Everyday Addition — Verifying $8 + 5 = 13$ After Solving on Paper

1. Type [8] and press [ENTER] to place 8 on Waiting Shelf Y. 2. Type [5] on Workbench X. 3. Press [+] to combine $Y + X = 13$!

Keys to Press: [8] [ENTER] [5] [+]

Example 2: Inside-Out Order of Operations Without Parentheses — $(8 + 4) \times (9 - 3) = 72$

1. Solve the first group: [8] [ENTER] [4] [+] → 12 waits visibly on Line Y! 2. Solve the second group: [9] [ENTER] [3] [-] → 6 sits on Line X. 3. Press [×] to multiply $12 \times 6 = 72$!

Keys to Press: [8] [ENTER] [4] [+] [9] [ENTER] [3] [-] [×]

Example 3: Real Fractions Using Two Dots — $1/3 + 1/4 = 7/12$

Pressing [.] twice types a real fraction ([.] [1] [.] [3] = 1/3). Press [ENTER], type 1/4 ([.] [1] [.] [4]), and press [+] to see 7/12!

Keys to Press: [•] [1] [•] [3] [ENTER] [•] [1] [•] [4] [+]

3 SOCRATIC QUESTIONS TO ASK AT THE KITCHEN TABLE (INSTEAD OF GIVING THE ANSWER)

1. "Before we check on the calculator, can you show me or sketch what this problem looks like with blocks?"

2. "Which inner part of this expression do we need to solve first on paper so it can wait on Shelf Y?"

3. "What quantity is sitting on Waiting Shelf Y right now, and what quantity is on Workbench X?"

Free printable homeschool schedules, block templates & answer keys: https://stackcalc.io/curriculum/answers

SUGGESTED 15-MINUTE WEEKLY HOMESCHOOL & FAMILY MATH LAB RHYTHM

• Mon–Thu (During Regular Math): Keep your core curriculum (Singapore, Saxon, etc.); use blocks & stack to self-check.

• Friday Lab (15–20 Min): Complete 1 unit section in Early/Upper Elementary Lab, Middle School Lab, or High School Lab.

Student Name: _____ **Parent / Coach Signature:** _____